How Should We Interpret Space Dimension?

A Trial for a Mathematical Foundation in Higher Dimensional Physics

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In memory of the late

Prof. Ichiro Yokota

of

Shinshu University

and

known for his research of cellular decompositions of classical Lie groups and realizations of exceptional Lie groups

Abstract

In modern physics we could say that space dimension is derived from physical conditions. Kaluza-Klein theory and D-brane are typical examples. However, not only by such conditions, we should also think about space dimension with insights from known facts without any physical conditions. In this talk we rethink space dimensionality from scratch.

1. Examples of Physical Conditions

- The standard model requires 19 numerical constants whose values are determined by experimental results. That is one of the motivations for particle physicists to establish a unified theory beyond the standard model.
- Hendrik Lorentz and George FitzGerald introduced a strange idea that space is 'contracted' to explain the results of the Michelson-Morley experiment: the Lorentz-FitzGerald contraction.
- Epicycles given on the orbits of planets to support geocentric theory. It made more accurate the geocentric theory than Copernicus's in the 16th century.
- Astronomers before Isaac Newton believed that the orbits of planets were on the celestial spheres.

- Universe models based on 3-sphere or 3-hyperboloid.

The problem needs to be explained in detail. Let us think of an easy example. There is a ball rolling on the table as shown in Figure 1 (Fig. 1). This could be considered a motion only on the plane if it does not bounce. So, we could say it is a motion in 2-dimensional space. On the other hand, could we really say that 'this is motion in 2-dimensional space'? This way of thinking is based on the concept of 'binding condition' by analytical dynamics. It ignores the vertical degree of freedom because the gravity of the ball is kept in balance with the normal force by the table. Therefore, it is a fake 2-dimensional motion.

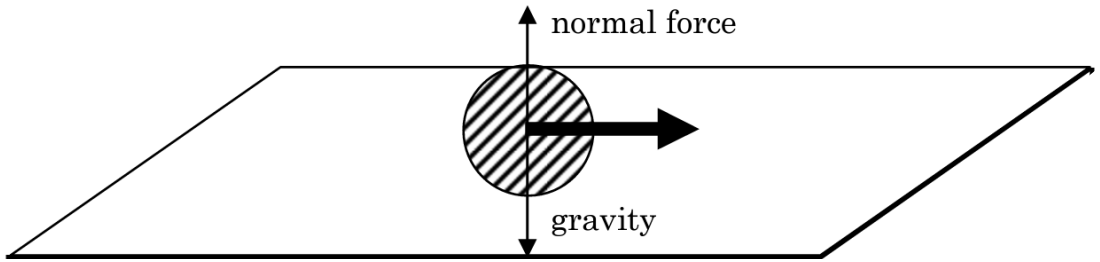


Fig.1

We can think of ourselves on Earth in this way, which means we are living in a closed 2-dimensional space or sphere. Furthermore, what about the thinking in cosmology that we are in a universe which is a '3-dimensional' sphere or 3-sphere in topology? It will be true that this is also based on the concept of 'binding condition'. If so, as shown in Fig. 2, the centrifugal force given to our galaxy moving in the Universe is perfectly balanced with a 'centripetal one'.

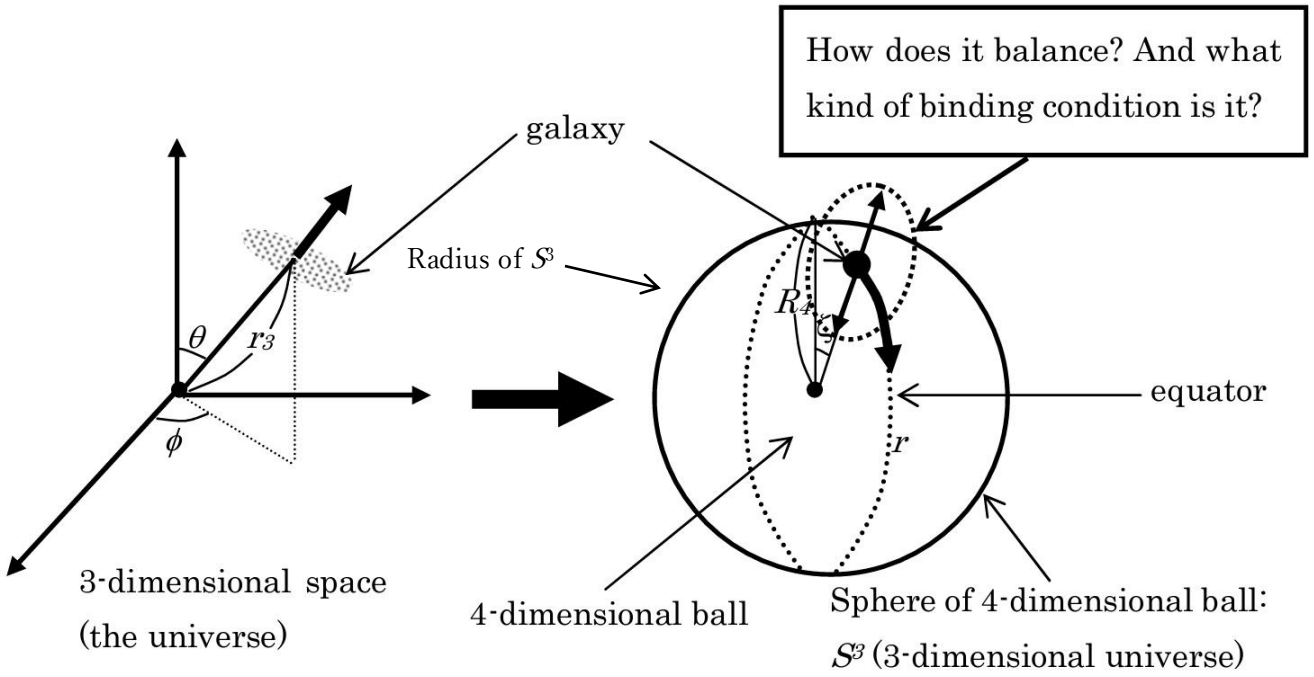


Fig. 2

Does this give a ‘binding condition’ to our galaxy just like a ball on the table and therefore we cannot perceive the 4th-dimensional degree of freedom? Especially, what is this kind of ’centripetal force’? Does that hint at a 5th force (quintessence) following gravity, electro-magnetic force, weak force and strong force in nature? Otherwise, as shown in Fig. 3, our galaxy will be thrown into the ‘4th-dimensional space’. Or there is something special to support the galaxy like the celestial sphere?

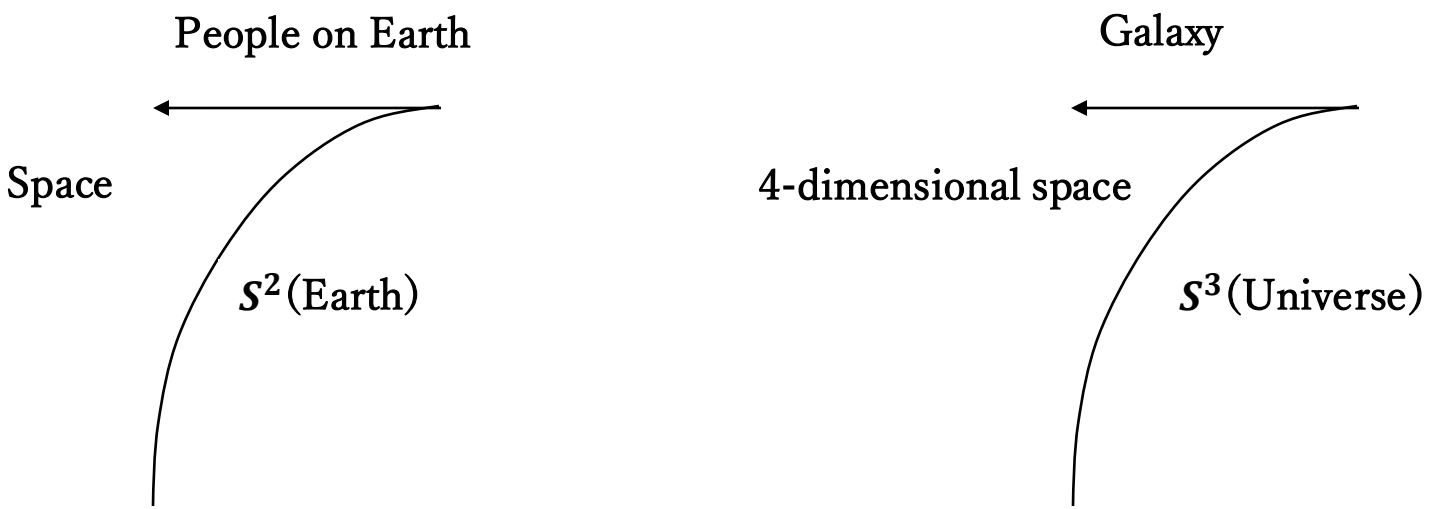


Fig. 3

As well as that, we would be thrown away into ‘space’ if not keeping the balance between the centrifugal force of the Earth’s rotation and gravity.

Thus, we are sure that introducing fewer physical conditions is preferable.

2. Concrete Insights

Given physical conditions, we historically knew that even wrong ideas were justified. Especially for higher dimensional space, we had to assume an unknown force. Kaluza-Klein theory also needs an unknown force to compactify extra-dimensional space as well as the 3-manifold universe model mentioned above, as it is like a microscopic celestial sphere. Therefore, it is important to know what the spatial dimension is. First of all, let us delve deeper into this matter.

Facts

- Fact 1. We have never detected any particles travelling or confined in a 2-dimensional space or plane forever in our universe. The case will be also about D2-brane in our 3-dimensional universe.
- Fact 2. Suppose for example, that we are living in 5-dimensional space but we only have 3 variables (x ,y, z). That is why we cannot see extra-dimensional space. However, this thought is wrong. This is clear from Fact 1.
- Fact 3. We are living on Earth on a 2-dimensional sphere. However, who amongst us believes that we are living in a 2-dimensional space?

Now, let us consider those facts above as an axiom.

Axiom 1

Lower dimensional spaces are not subspaces of higher dimensional ones. They are disjoint of each other as shown in Fig. 4.

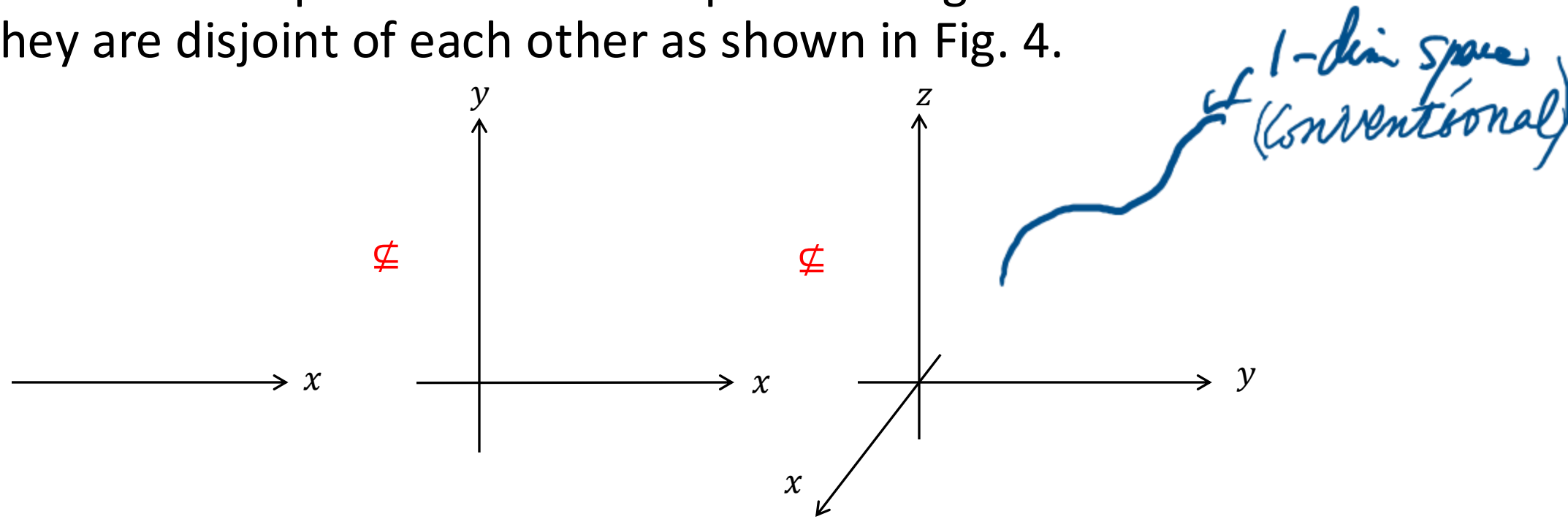


Fig. 4

They are different worlds from each other. Let us imagine a situation like that of Alice in Wonderland. She starts from our 3-dimensional world, where the point has 'theoretically' the 3 variables of x, y, and z, or in other words 3 degrees of freedom, x, y, and z. She, in our 3-dimensional space is transferred to an 1-dimensional tunnel. In the tunnel, will she have only one degree of freedom?

When Alice returns from the tunnel to the usual 3-dimensional world, what will become of the variables or the degree of freedom that she has?

The answer is that Alice has the 3 variables of x, y, z even in 1-dimensional space. Even if she goes to a much higher dimensional world like 10-dimensional space, she has only the 3 variables or co-ordinates of x, y, z; though that does NOT mean (x, y, z, 0, 0, 0, 0, 0, 0, 0). Then when she comes back to our world, it is all the same as before she travelled into 1-dimensional space. Let us consider these matters.

A point mapped from lower dimensional space to higher dimensional space

An equation $x = a$ is a point in 1-dimensional space, line in 2-dimensional space and plane in 3-dimensional space. Though the equation is always the same, the graph is different among different dimensional spaces as shown in Fig. 5. We should discuss them from a unified viewpoint.

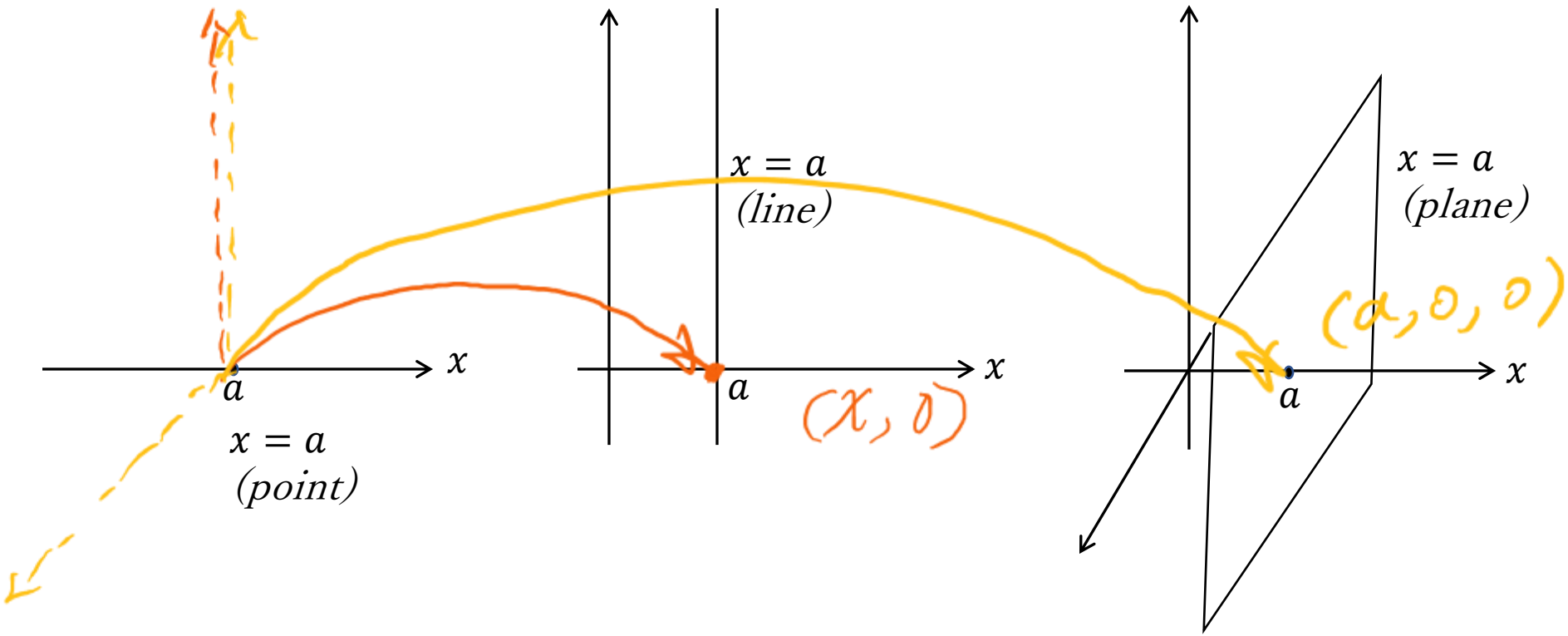


Fig. 5

Definition 1

- If point $x = a$ in 1-dimensional space is mapped to any higher dimensional spaces, y and z co-ordinates or variables for the point are never given.

For example, if $x = a$ in 1-dimensional space is mapped into 2-dimensional space, the co-ordinate keeps $x = a$, NOT $(x, y) = (a, 0)$.

If $x = a$ in 1-dimensional space is mapped into 3-dimensional space, the co-ordinate keeps $x = a$, NOT $(x, y, z) = (a, 0, 0)$.

See also Fig. 6.

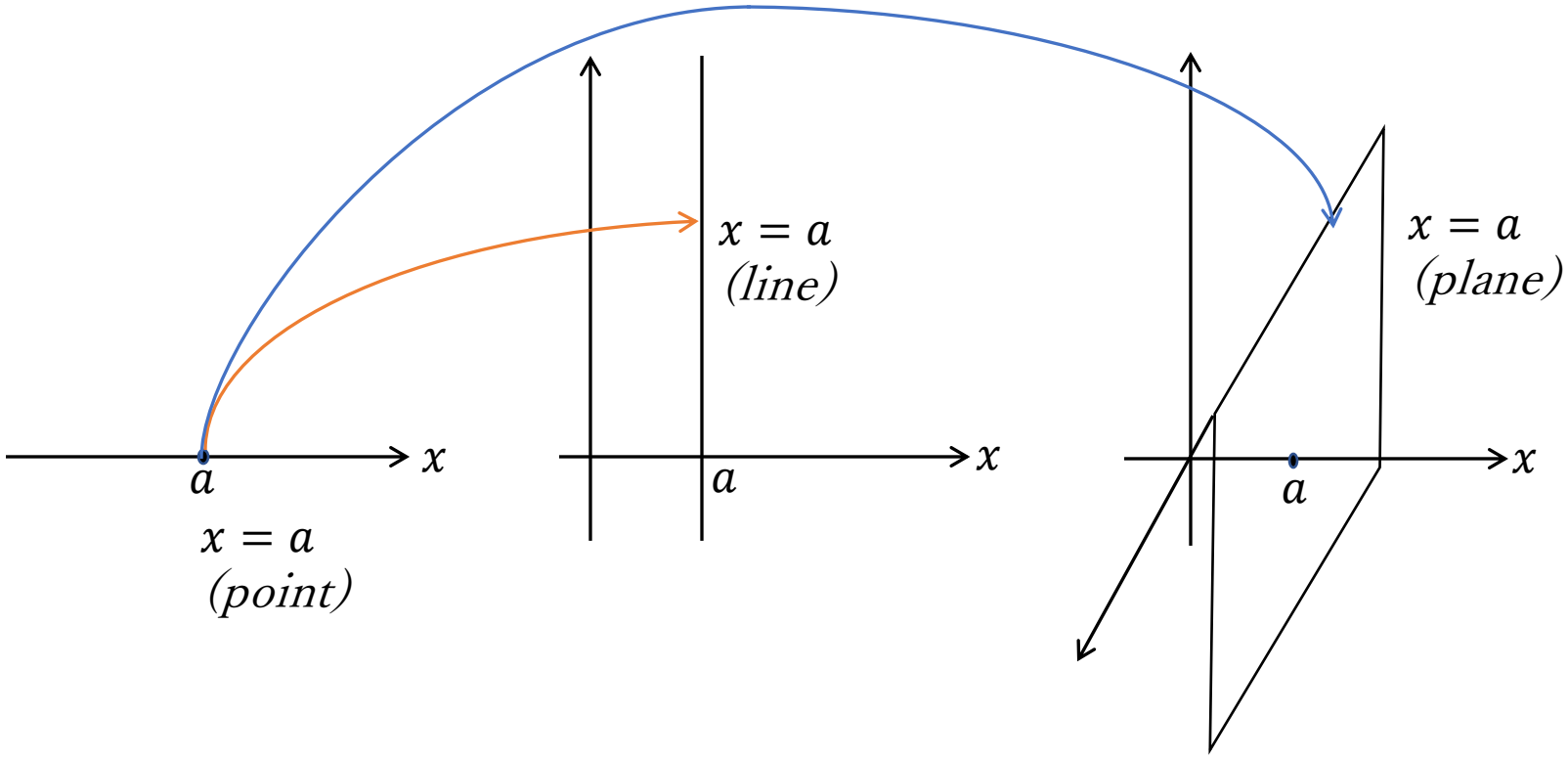


Fig. 6

Considering the practical case of projection into higher dimension, we should reconsider our thoughts about the system of equations from a graphical viewpoint. For example, let us consider a system of equations $f(x, y, z) = x^2 + y^2 + z^2 - 1 = 0$ and $\mathrm{g}(x, y) = x - y = 0$. Though it produces two equations $h_1(x, z) = 2x^2 + z^2 - 1 = 0$ and

$h_2(y, z) = 2y^2 + z^2 - 1 = 0$ simultaneously; it seems not to make sense because these equations of ellipse seem not to be solutions indicating intersections of the sphere $f(x, y, z)$ and the line $g(x, y)$. However, considering $g(x, y)$ as a plane projected into 3-dimensional space, it makes sense that $h_1(x, z)$ and $h_2(y, z)$ are equations which are projected into x-z plane and y-z plane, as shown in Fig. 7.

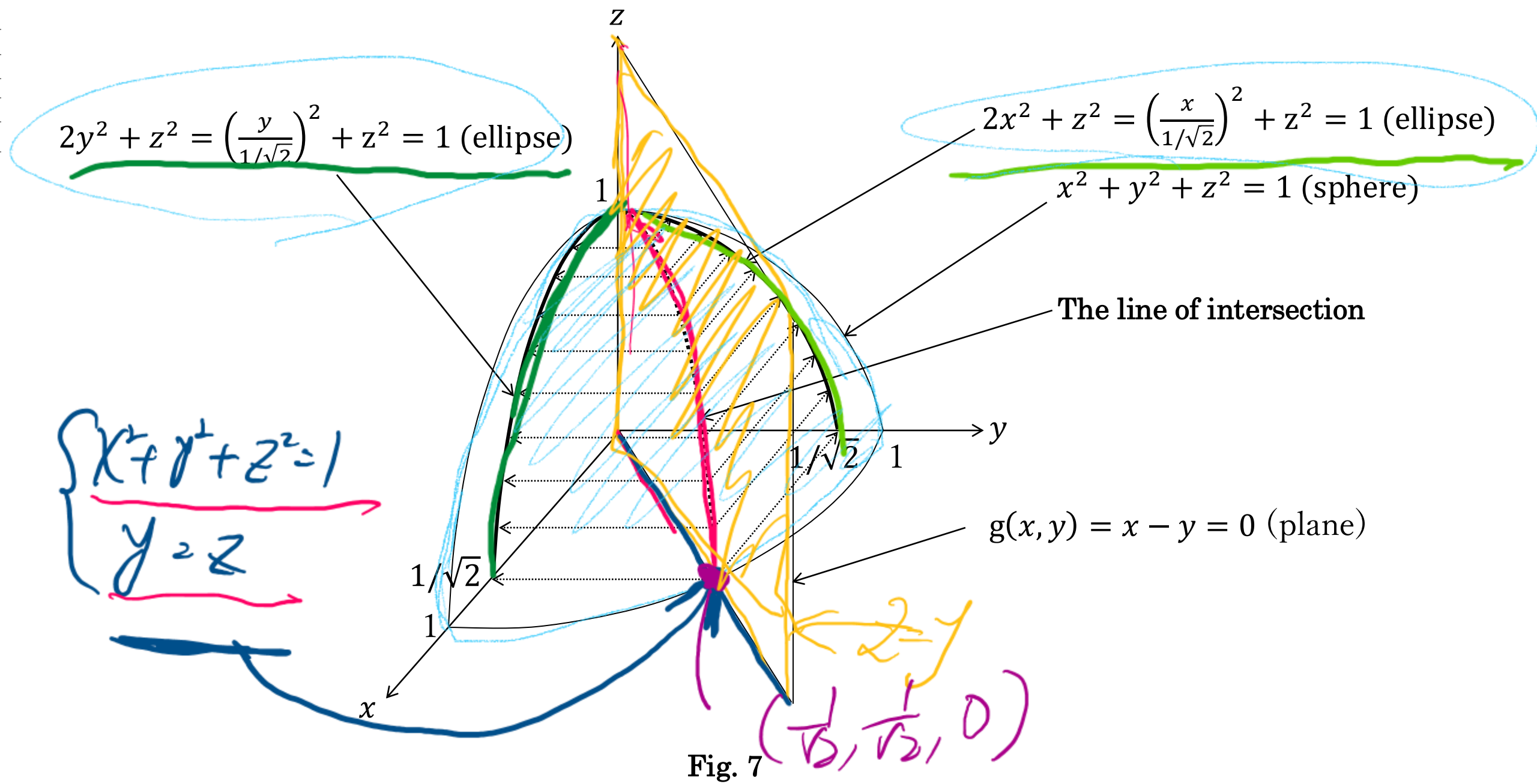

$2y^2 + z^2 = \left(\frac{y}{1/\sqrt{2}}\right)^2 + z^2 = 1$ (ellipse)
$2x^2 + z^2 = \left(\frac{x}{1/\sqrt{2}}\right)^2 + z^2 = 1$ (ellipse)
$x^2 + y^2 + z^2 = 1$ (sphere)
The line of intersection
$g(x, y) = x - y = 0$ (plane)
z
y
x
1
$1/\sqrt{2}$
1
$1/\sqrt{2}$
1
$\begin{cases} x^2+y^2+z^2=1 \\ y=z \end{cases}$
$z=y$
$\left(\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}, 0\right)$

Fig. 7

Since the curve of intersection is indicated by $(x, y, z) = (h_1(x, z), h_2(y, z), z)$ as a plane projected into 3-dimensional space,

$$\therefore (x, y, z) = \left(\pm\sqrt{\frac{1-z^2}{2}}, \pm\sqrt{\frac{1-z^2}{2}}, z\right), \because -1 \leq z \leq 1.$$

Remark. From the viewpoint of quantum theory, it might suggest a wave function collapse.

Only if $z = 0$ (plane), then $(x, y, z) = \left(\pm\sqrt{1/2}, \pm\sqrt{1/2}, 0\right)$. It shows that the sphere $(x, y, z) = x^2 + y^2 + z^2 - 1 = 0$ intersects the planes $x - y = 0$ and $z = 0$.

Based on the result above, the system of arbitrary equations below could make sense:

$$\begin{cases} f_1(x_1, x_2, x_3, \ldots\ldots, x_l) = 0 \\ f_2(x_1, x_2, x_3, \ldots\ldots\ldots\ldots, x_m) = 0 \\ \vdots \\ f_k(x_1, x_2, x_3, \ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots, x_r) = 0 \end{cases}$$

A Point transferred from Lower Dimensional Space to Higher Dimensional Space

As well as mapping a point from lower dimensional space to higher dimensional space, we would like to define the map of a point from higher to lower dimensional space.

Definition 2

- If a point is mapped to lower dimensional space, the original number of co-ordinates never changes.

In other words, if a point is mapped from higher dimensional space to any of lower dimensional spaces, the degree of freedom or the number of its co-ordinates is never decreased.

Example 1

Let's consider a transformation of homogeneous co-ordinates. As shown in Fig. 8, the number of variables is 3: (x, y, z). On the other hand, as shown in Fig. 9, let us transform the co-ordinates via scaling or mapping f: $(x, y, z) \mapsto (x/z, y/z)$ and the inverse ϕ: $(x/z, y/z) \mapsto (x, y, z)$. Then the equation $x^2 + y^2 = z^2$ changes to $\left(\frac{x}{z}\right)^2 + \left(\frac{y}{z}\right)^2 = X^2 + Y^2 = 1$.

The relationship between $x^2 + y^2 = z^2$ in the 3-dimensional space and $\left(\frac{x}{z}\right)^2 + \left(\frac{y}{z}\right)^2 = X^2 + Y^2 = 1$ in 2-dimensional space is that the hyperboloid $x^2 + y^2 = z^2$ is projected on to the circle $\left(\frac{x}{z}\right)^2 + \left(\frac{y}{z}\right)^2 = X^2 + Y^2 = 1$.

HOWEVER, $\left(\frac{x}{z}\right)^2 + \left(\frac{y}{z}\right)^2 = 1$ 'keeps the number of variables '3'' in '2'-dimensional space: two true variables x and y, and the parameter z, which controls the radius of the circle. It is shown as a UNIT circle in Fig. 8.

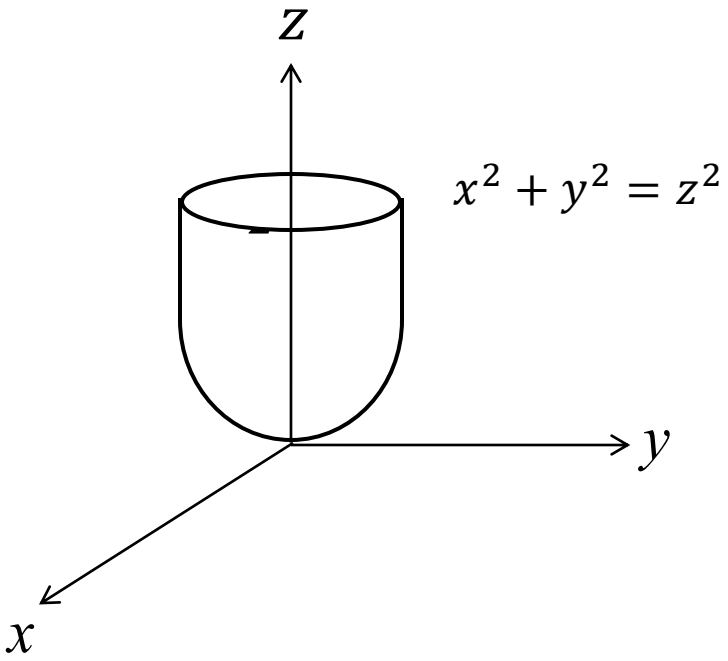

z
$x^2 + y^2 = z^2$
y
x

Fig. 8

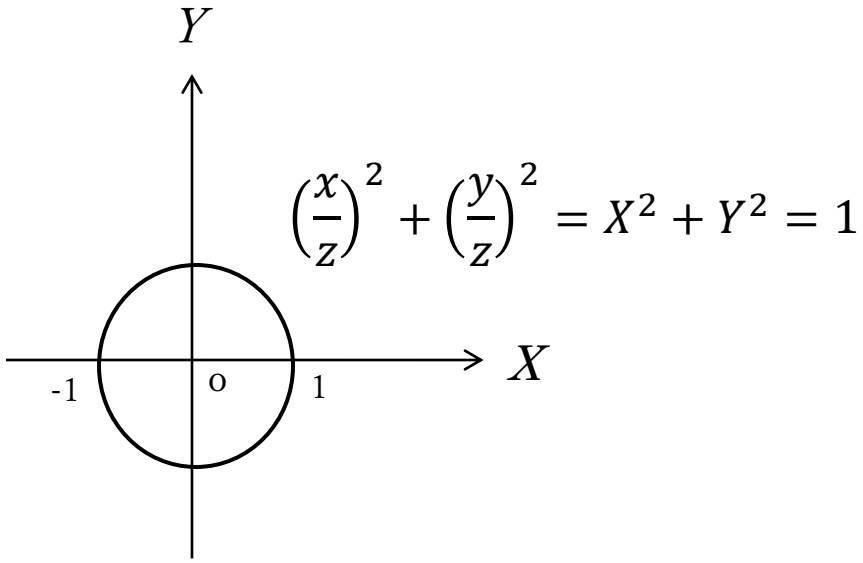

Y
$\left(\frac{x}{z}\right)^2 + \left(\frac{y}{z}\right)^2 = X^2 + Y^2 = 1$
X
-1
o
1

Fig. 9

Example 2

This is not the best example though. The case below is in a series of projections from higher to lower dimensional space. As shown in Fig. 10, a co-ordinate in $B^3 = (r_3, \alpha, \beta)$ is projected in $B^2 = (r_3 cos\alpha, \beta) = (r_2, \beta)$, B^2 in $B^1 = r_3 cos\alpha cos\beta = r_2 cos\beta = r_1$, B^3 in B^1.

$\therefore$ $B^3 \rightarrow B^2$ $\quad$ $B^2 \rightarrow B^1$ $\quad$ $B^3 \rightarrow B^1$

$(r_3, \alpha, \beta) \mapsto (r_2, \beta)$ $\quad$ $(r_2, \beta) \mapsto r_1$ $\quad$ $(r_3, \alpha, \beta) \mapsto r_1$

The commutative diagram with morphisms ($f \circ g = E_{32}E_{21} = E_{31}$) is shown in Fig. 11.

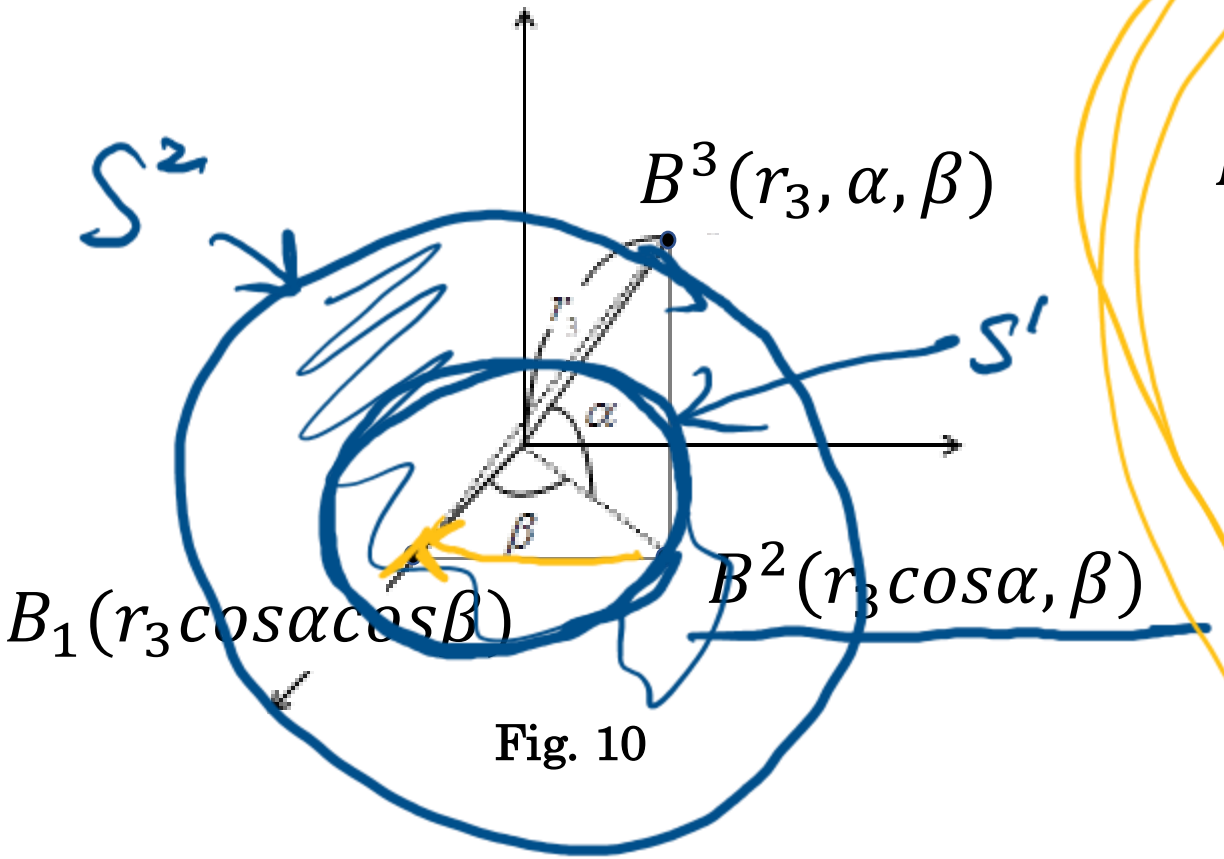


Fig. 10

morphism

$$\begin{tikzcd} B^3 \arrow[r, "E_{32}"] \arrow[dr, "E_{31}"'] & B^2 \arrow[d, "E_{21}"] \arrow[dl, phantom, "\circlearrowright"] \\ & B^1 \end{tikzcd}$$

Fig. 11

Likewise, mapping into lower dimensional space by this method, the commutative diagram with morphisms (Fig. 11) can be changed to any other combinations by:

$(r_n, \theta_1, \dots, \theta_{n-1}) \in B^n$,

$(r_{n-1}(= r_n cos\theta_n), \theta_1, \dots, \theta_{n-1}) \in B^{n-1}$,

$(r_{n-2}(= r_n cos\theta_n cos\theta_{n-1}), \theta_1, \dots, \theta_{n-2}) \in B^{n-2}$,

.....

.....

.....

$(r_2(= r_n cos\theta_n cos\theta_{n-1} \dots cos\theta_2), \theta_1) \in B^2$,

$r_1(= r_n cos\theta_n cos\theta_{n-1} \dots cos\theta_1) \in B^1$.

Example 3

We utilize the fundamental theorem on homomorphisms for linear algebra. Let *V* and *W* be vector spaces, *f* be a linear mapping from *V to W*, then we have

$$V/\mathrm{Ker}(f) \simeq \mathrm{Im}(f)\ .\ \ldots (2.1)$$

Let *V* and *W* be vector spaces, *f* be a linear mapping from *V to W*,

$f: \mathrm{R}^3 \ni \begin{pmatrix} x \\ y \\ z \end{pmatrix} \mapsto x + y + z \in \mathrm{R}$ is a homomorphism.

$$\because f(g_1 + g_2) = f\left(\begin{pmatrix} x_1 \\ y_1 \\ z_1 \end{pmatrix} + \begin{pmatrix} x_2 \\ y_2 \\ z_2 \end{pmatrix}\right) = f\begin{pmatrix} x_1 + x_2 \\ y_1 + y_2 \\ z_1 + z_2 \end{pmatrix}$$

$$= (x_1 + x_2) + (y_1 + y_2) + (z_1 + z_2) = f\begin{pmatrix} x_1 \\ y_1 \\ z_1 \end{pmatrix} + f\begin{pmatrix} x_2 \\ y_2 \\ z_2 \end{pmatrix}$$

$= f(g_1) + f(g_2)$.

Here, $g_i \in (x_i, y_i, z_i, \ldots)$.

However, since the bijection of f is not verified, we cannot refer to the inverse. Instead, we consider $\bar{\boldsymbol{f}} : G/K(f) \to G'$.

Paying attention to $\begin{pmatrix} x \\ y \\ z \end{pmatrix} = x\begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix} + y\begin{pmatrix} 0 \\ 1 \\ -1 \end{pmatrix} + \begin{pmatrix} 0 \\ 0 \\ x+y+z \end{pmatrix}, \begin{pmatrix} 0 \\ 0 \\ x+y+z \end{pmatrix} + \mathrm{R}\begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix} + \mathrm{R}\begin{pmatrix} 0 \\ 1 \\ -1 \end{pmatrix} =$

$\begin{pmatrix} x \\ y \\ z \end{pmatrix} + \mathrm{R}\begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix} + \mathrm{R}\begin{pmatrix} 0 \\ 1 \\ -1 \end{pmatrix}$.

Since $K(f) = \left\{\begin{pmatrix} x \\ y \\ z \end{pmatrix}, f\begin{pmatrix} x \\ y \\ z \end{pmatrix} = 0\right\} = \left\{\begin{pmatrix} x \\ y \\ z \end{pmatrix}, x+y+z = 0\right\}$,

$\bar{f}: \mathrm{R}^3/K(f) = \mathrm{R}^3/\left(\mathrm{R}\begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix} + \mathrm{R}\begin{pmatrix} 0 \\ 1 \\ -1 \end{pmatrix}\right) = \left\{\begin{pmatrix} x \\ y \\ z \end{pmatrix} + \mathrm{R}\begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix} + \mathrm{R}\begin{pmatrix} 0 \\ 1 \\ -1 \end{pmatrix} : \begin{pmatrix} x \\ y \\ z \end{pmatrix} \in \mathrm{R}^3\right\} \mapsto x+y+z \in \mathrm{R}.$

See also Fig.12.

Since $x+y+z$ has 3 degrees of freedom or 3 variables in R, the invariant is 3 .

$$(0,0,s+t+u)^T + (s,0,-s)^T + (0,t,-t)^T = (0,0,s+t+u)^T + (s,t,-s-t)^T = (s,t,u)^T$$

$$K(f) = (s,0,-s)^T + (0,t,-t)^T (= (s,t,-s-t)^T), f(s,t,-s-t)^T = 0$$

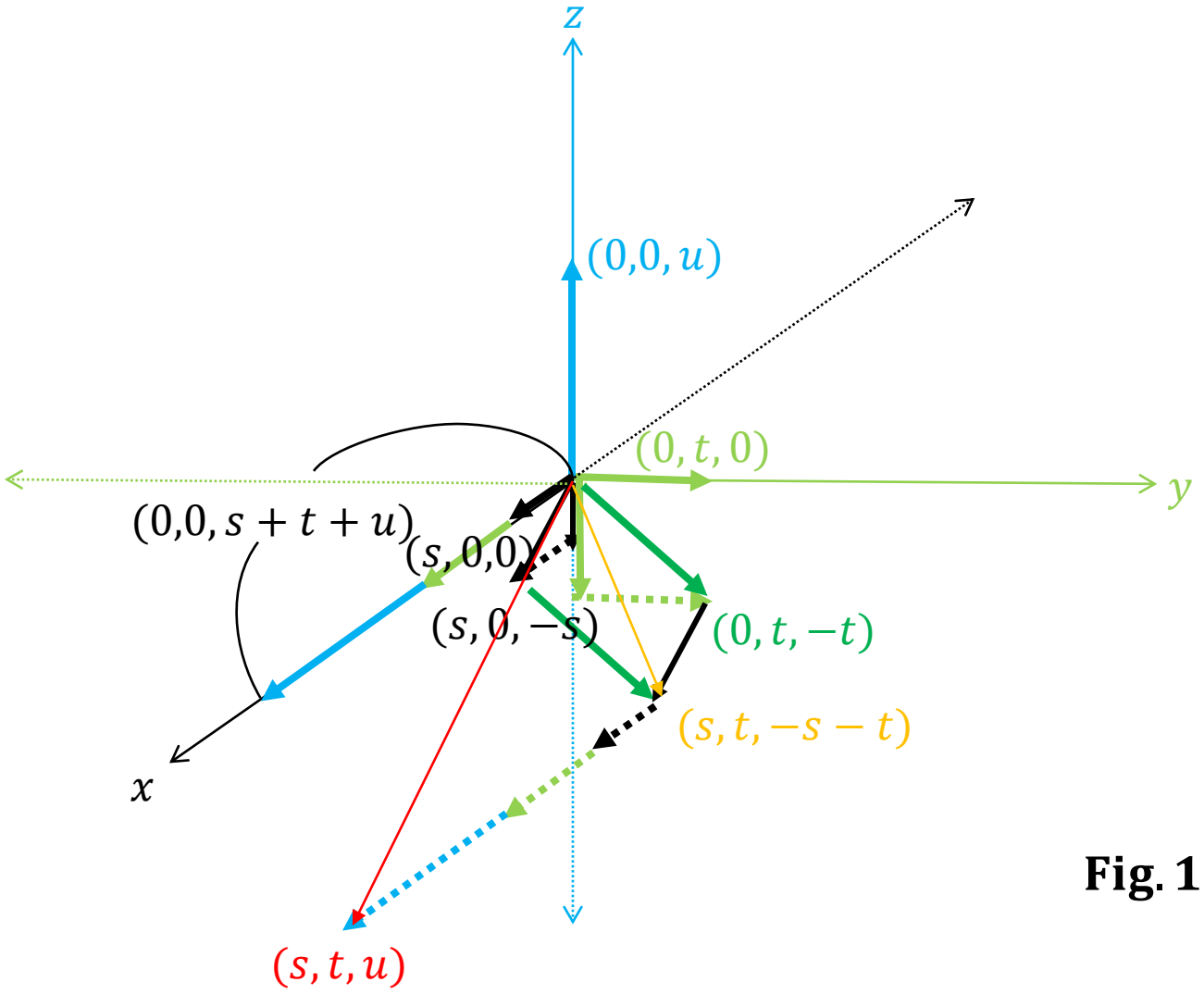


Fig. 12

Remark. $\begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix}$ and $\begin{pmatrix} 0 \\ 1 \\ -1 \end{pmatrix}$ are perpendicular to $\begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$,
$\because\ (1,0,-1)(1,1,1,) = (0,1,-1)(1,1,1) = 0$. The orthogonality is significant in this discussion.

The inverse ϕ is as follows. Since ϕ: $\mathrm{Image} f \ni h = f(g) \to gK \in G/K$,

$$\mathrm{R} \ni \bar{f}(g) = x + y + z \mapsto \mathrm{R}^3 / \left(\mathrm{R}\begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix} + \mathrm{R}\begin{pmatrix} 0 \\ 1 \\ -1 \end{pmatrix} \right) = \left\{ \begin{pmatrix} x \\ y \\ z \end{pmatrix} + \mathrm{R}\begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix} + \mathrm{R}\begin{pmatrix} 0 \\ 1 \\ -1 \end{pmatrix} : \begin{pmatrix} x \\ y \\ z \end{pmatrix} \in \mathrm{R}^3 \right\}.$$

Now, let us pay attention to the invariant or symmetry throughout a series of the mappings above.

For example, let us think of $f\colon \mathrm{R}^4 \ni \begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{pmatrix} \mapsto \begin{pmatrix} x_1 + x_3 \\ x_2 + x_4 \end{pmatrix} \in R^2$. Or another case $\mapsto \begin{pmatrix} x_1 + x_3 + x_4 \\ x_2 \end{pmatrix}$ is also fine.

Then, $K(f) = \left\{ \begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{pmatrix}; f\begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{pmatrix} = 0 \right\} = \left\{ \begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{pmatrix}; \begin{pmatrix} x_1 + x_3 \\ x_2 + x_4 \end{pmatrix} = 0 \right\} = \left\{ s\begin{pmatrix} 1 \\ 0 \\ -1 \\ 0 \end{pmatrix} + t\begin{pmatrix} 0 \\ 1 \\ 0 \\ -1 \end{pmatrix}; s, t \in R \right\} = \left\{ R\begin{pmatrix} 1 \\ 0 \\ -1 \\ 0 \end{pmatrix} + R\begin{pmatrix} 0 \\ 1 \\ 0 \\ -1 \end{pmatrix} \right\}$.

Since $\mathrm{R}^4/K(f) = \mathrm{R}^4/\left\{ R\begin{pmatrix} 1 \\ 0 \\ -1 \\ 0 \end{pmatrix} + R\begin{pmatrix} 0 \\ 1 \\ 0 \\ -1 \end{pmatrix} \right\} = \left\{ \begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{pmatrix} + R\begin{pmatrix} 1 \\ 0 \\ -1 \\ 0 \end{pmatrix} + R\begin{pmatrix} 0 \\ 1 \\ 0 \\ -1 \end{pmatrix}; \begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{pmatrix} \in \mathrm{R}^4 \right\}$,

$\bar{f}(g) = E_{42}x = \mathrm{R}^4/K(f) \mapsto \begin{pmatrix} x_1 + x_3 \\ x_2 + x_4 \end{pmatrix}$.

Remark. For $\begin{pmatrix} 1 \\ 1 \\ 1 \\ 1 \end{pmatrix} \in R^4$, $\begin{pmatrix} 1 \\ 0 \\ -1 \\ 0 \end{pmatrix}$ and $\begin{pmatrix} 0 \\ 1 \\ 0 \\ -1 \end{pmatrix}$ are orthogonal. Thinking of another mapping based on the orthogonality,

We are naturally able to think of the case $f\colon \mathrm{R}^5 \ni \begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \end{pmatrix} \mapsto \begin{pmatrix} x_1 + x_3 + x_5 \\ x_2 + x_4 \end{pmatrix} \in R^2$.

Then, $K(f) = \left\{ \begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \end{pmatrix} ; f \begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \end{pmatrix} = 0 \right\} = \left\{ \begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \end{pmatrix} ; \begin{pmatrix} x_1 + x_3 + x_5 \\ x_2 + x_4 \end{pmatrix} = 0 \right\} = \left\{ s \begin{pmatrix} 1 \\ 0 \\ -1 \\ 0 \\ 0 \end{pmatrix} + t \begin{pmatrix} 0 \\ 1 \\ 0 \\ -1 \\ 0 \end{pmatrix} + u \begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \\ -1 \end{pmatrix} =; s, t, u \in R \right\} = \left\{ R \begin{pmatrix} 1 \\ 0 \\ -1 \\ 0 \\ 0 \end{pmatrix} + R \begin{pmatrix} 0 \\ 1 \\ 0 \\ -1 \\ 0 \end{pmatrix} + R \begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \\ -1 \end{pmatrix} \right\}$.

Since $\mathrm{R}^5/K(f) = \mathrm{R}^5 / \left\{ R \begin{pmatrix} 1 \\ 0 \\ -1 \\ 0 \\ 0 \end{pmatrix} + R \begin{pmatrix} 0 \\ 1 \\ 0 \\ -1 \\ 0 \end{pmatrix} + R \begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \\ -1 \end{pmatrix} \right\} = \left\{ \begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \end{pmatrix} + R \begin{pmatrix} 1 \\ 0 \\ -1 \\ 0 \\ 0 \end{pmatrix} + R \begin{pmatrix} 0 \\ 1 \\ 0 \\ -1 \\ 0 \end{pmatrix} + R \begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \\ -1 \end{pmatrix} ; \begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \end{pmatrix} \in \mathrm{R}^5 \right\}$,

$\bar{f}(g) = E_{52}x = \mathrm{R}^5/K(f) \mapsto \begin{pmatrix} x_1 + x_3 + x_5 \\ x_2 + x_4 \end{pmatrix}$.

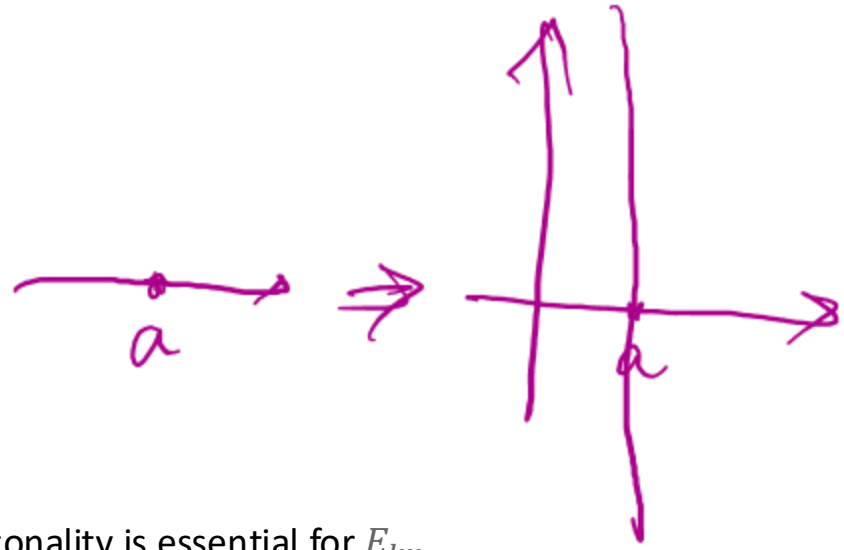
Thus, we can see the law-like composition for $\mathrm{R}^n/K(f)$.

On the other hand, let us think of mapping from the lower dimensional space to a higher one.

For example, for $E_{13}: \mathrm{R}^1 \to \mathrm{R}^3/k(f)$,

$x \mapsto \begin{pmatrix} x \\ 0 \\ 0 \end{pmatrix} + \mathrm{R} \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} + \mathrm{R} \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} = \begin{pmatrix} x \\ * \\ * \end{pmatrix}$. See also Fig. 13.

$x = a$

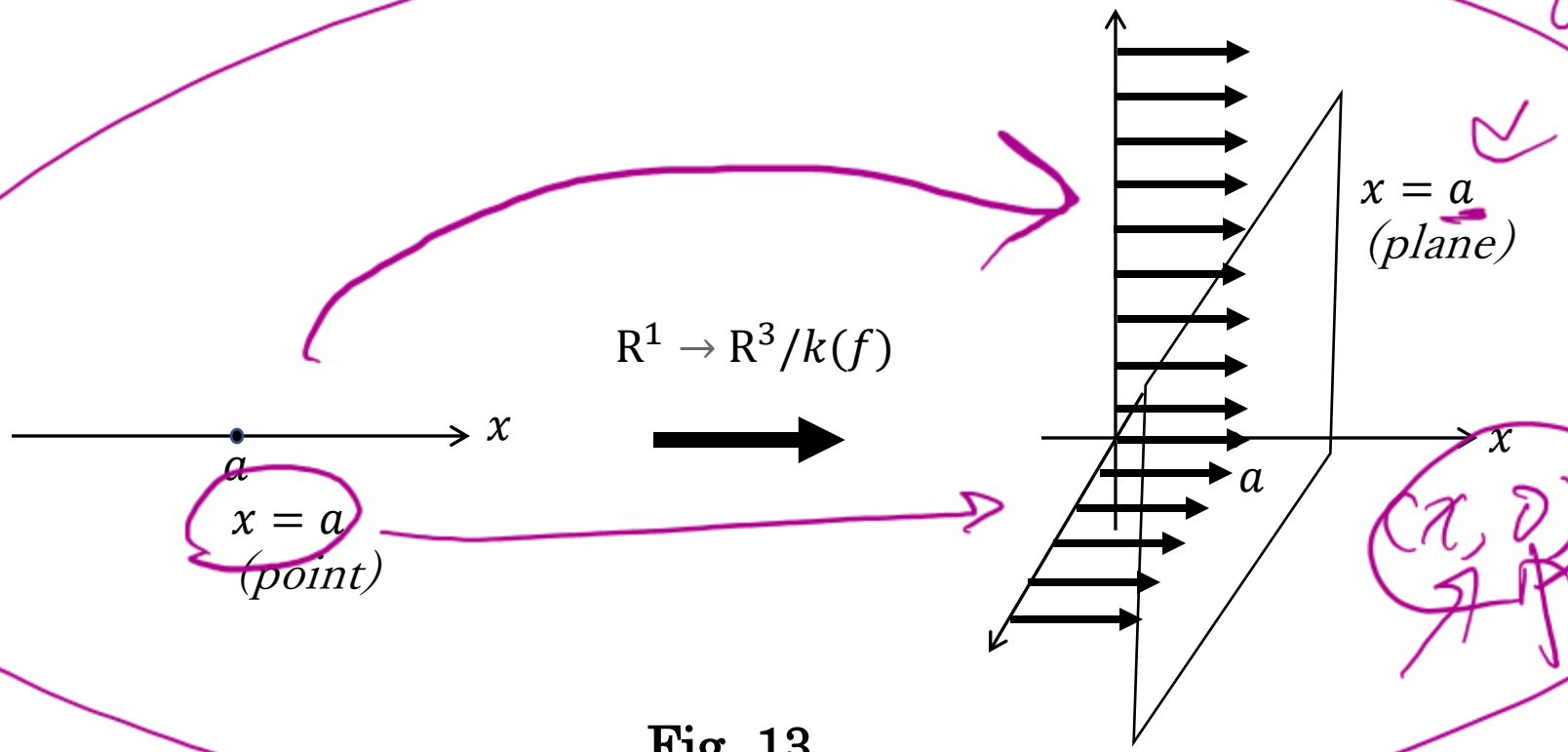


In this case, the unite vector $\begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}$ of $\begin{pmatrix} x \\ 0 \\ 0 \end{pmatrix}$ is also orthogonal to $\begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}$ and $\begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$. Thus, the orthogonality is essential for E_{lm}.

$$x = a \mapsto \begin{pmatrix} a \\ 0 \\ 0 \end{pmatrix} + \mathrm{R}\begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} + \mathrm{R}\begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} = \begin{pmatrix} x \\ * \\ * \end{pmatrix}$$

1-p 2-p

one dof.



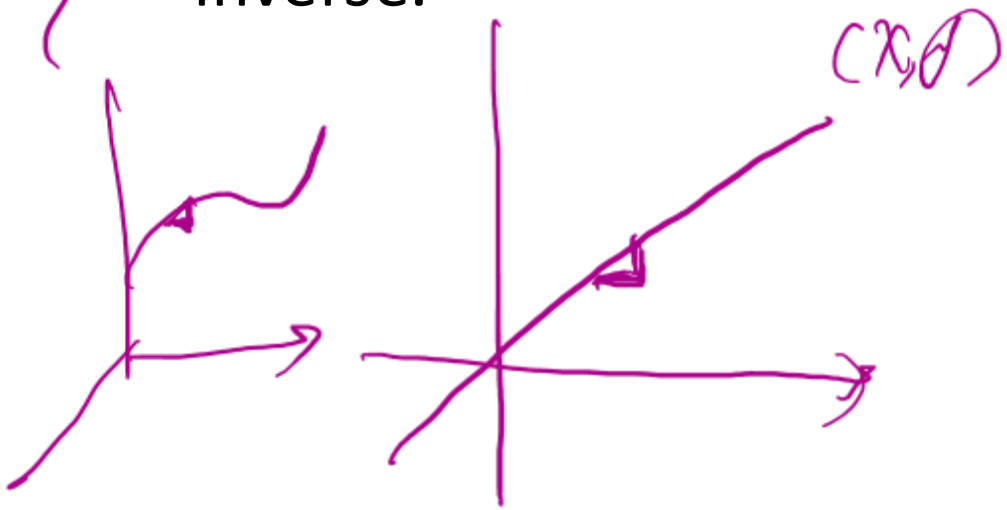
(x, 0)

2 degrees of freedom

Fig. 13

Example 4

In the context of dimensional reduction, we could think of differentiation for some polynomial equations. For example, $\frac{f(x,y)}{\partial x \partial y} = z_{xy}$ is also considered a mapping from 3-dimensional to 1-dimensional space. It is natural that we can consider integration as the inverse.



$$f'(x) = \frac{dy}{dx} =$$

Remark

In transferring any point among different dimensional spaces, the original number of variables of any point remains the same.

Those propositions above can be wrapped up in another axiom:

Axiom 2

The number of variables of any point in the original n-dimensional space, remains the same if the point transfers into any other dimensional space. It is equivalent to an invariant of topology.

3. Description with Matrices

Let us introduce a specific matrix operator to project a point between mutual dimensions. This matrix is different from a conventional one. It includes a special operator needing a temporary variable for operation. This is because the number of variables of a point before and after this operation is different. Demonstrating it with attention to the fact, for example, operating by an operator E_{12} to map a point A_1 in the 1-dimensional space into the 2-dimensional space (A_2), the equation is $A_2 = E_{12}A_1$,

$$\therefore \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & D \end{pmatrix} \begin{pmatrix} a \\ T \end{pmatrix} = \begin{pmatrix} a \\ DT \end{pmatrix}, \ldots (3.1)$$

where *D* is a matrix element making the dimension higher and *T* a temporary variable to correspond to the 2-dimensional space after operation. Therefore, *DT* denotes all real numbers of *y* at once. This process of Eq. 3.1 is shown in Fig. 14.

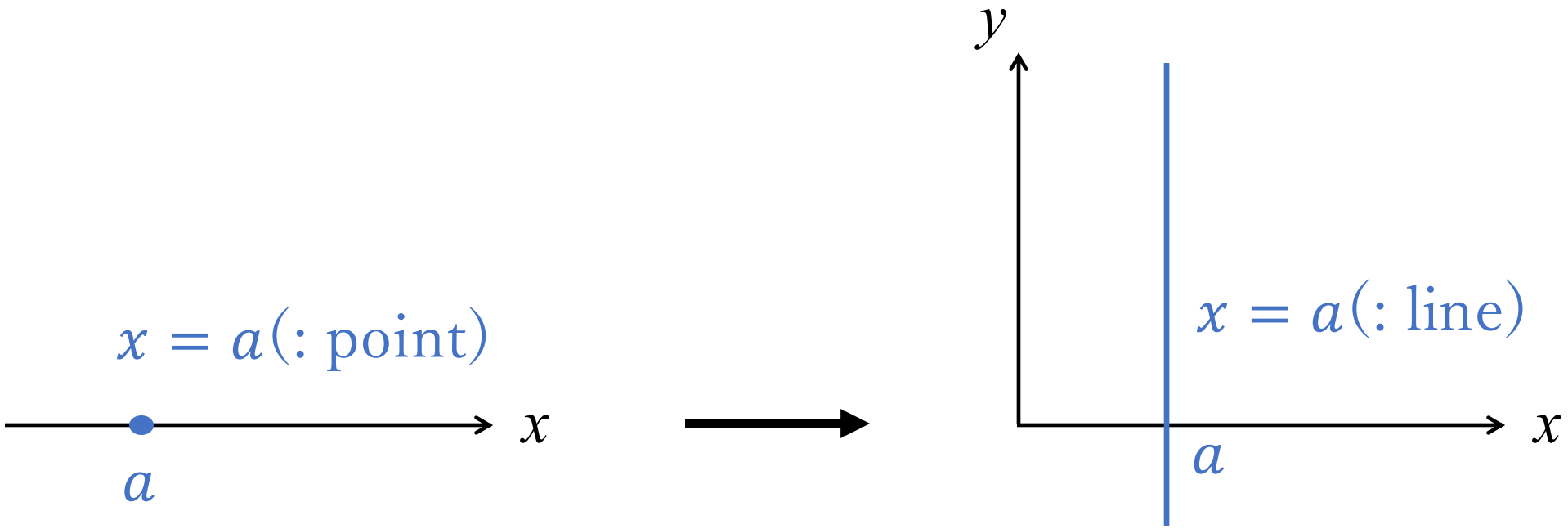


Fig. 14

Operating another case from 1-dimensional to 3-dimensional space, then:

$$\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & D & 0 \\ 0 & 0 & D \end{pmatrix} \begin{pmatrix} a \\ T \\ T \end{pmatrix} = \begin{pmatrix} a \\ DT \\ DT \end{pmatrix}. \quad \ldots (3.2)$$

Similarly, projection from 3-dimensional to 2-dimensional space is operated as:

$$\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & D^{-1} \end{pmatrix} \begin{pmatrix} a \\ b \\ c \end{pmatrix} = \begin{pmatrix} a \\ b \\ D^{-1}c \end{pmatrix}. \quad \ldots (3.3)$$

D^{-1} denotes an element making the dimension lower and inverse of D. Eq. 3.3 is shown in Fig. 15. Then, if returning the point projected from 3-dimensional into 2-dimensional space by Eq. 3.3 to the original dimensional space, the operation is as follows:

$$\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & D \end{pmatrix} \begin{pmatrix} a \\ b \\ D^{-1}c \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & D \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & D^{-1} \end{pmatrix} \begin{pmatrix} a \\ b \\ c \end{pmatrix} = \begin{pmatrix} a \\ b \\ c \end{pmatrix}. \quad \ldots (3.4)$$

$$\therefore E_{32}E_{23}(= (E_{23})^{-1}E_{23}) = E_{33} \equiv 1. \quad \ldots (3.5)$$

Remark. The operators E_{32} and E_{23} of Eq. 3.5 are ordered from left to right for convenience. The matrices of Eq. 3.4 are contrarily ordered for the operators and obviously calculated from right to left.

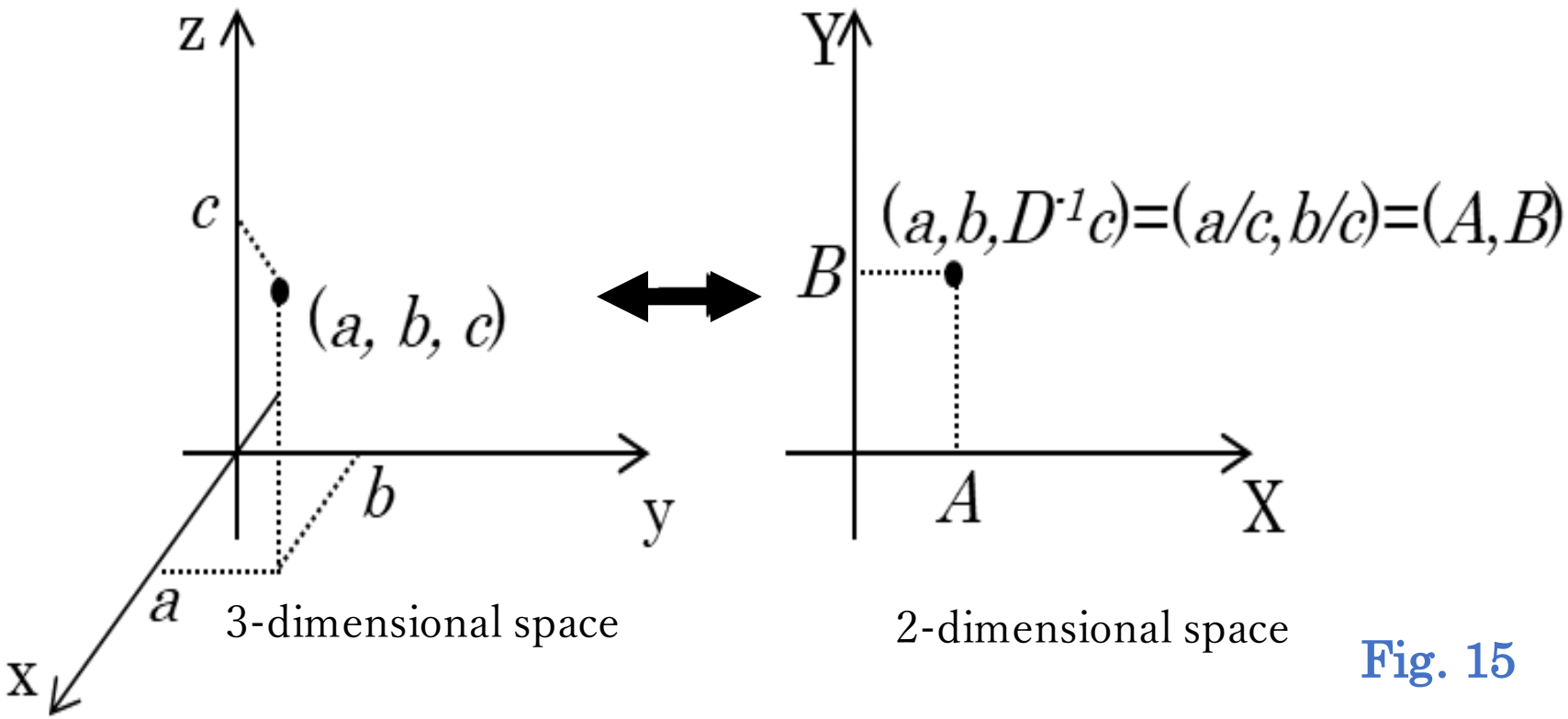

z
c
(a, b, c)
b
y
a
x
3-dimensional space
Y
(a,b,D-1c)=(a/c,b/c)=(A,B)
B
A
X
2-dimensional space

Fig. 15

The general operator which is dimensional unit matrix E_{lm} is:

If $l < m$,

$$E_{lm} = diag(\overbrace{1,1,1,\dots,1,1,1}^{l}, \overbrace{D,D,D,\dots,D,D,D}^{m-l}).$$

If $m < l$,

$$E_{lm} = (E_{ml})^{-1} = diag(\overbrace{1,1,1,\dots,1,1,1}^{m}, \overbrace{D^{-1},D^{-1},D^{-1},\dots,D^{-1},D^{-1},D^{-1}}^{l-m}).$$

$$\therefore E_{jk}E_{kj} = E_{jj} \equiv 1 \equiv E_{kj}E_{jk} = E_{kk}. \quad \dots (3.6)$$

Furthermore,

$$E_{0n} = diag(\overbrace{D,D,D,\dots,D,D,D}^{n}),$$

$$E_{n0} = (E_{0n})^{-1} = diag(\overbrace{D^{-1},D^{-1},D^{-1},\dots,D^{-1},D^{-1},D^{-1}}^{n}).$$

4. The Groupoid

The unchangeable number of variables can be considered as a 'symmetry' in physics or 'invariant' in mathematics. To make this point clearer we would like to discuss it in an algebraic way.

Proposition. *In the former chapter, matrices in a series of partially functional operations make the groupoid action. They are indicated by equations as follows*:

i. $E_{lm}E_{mn} = E_{ln}$ (automorphism, proven in Section 5), … (4.1)

ii. $(E_{kl}E_{lm})E_{mn} = E_{kl}(E_{lm}E_{mn})$ (associative), … (4.2)

iii. $E_{jk}E_{kj} = E_{jj} \equiv 1 \equiv E_{kj}E_{jk} = E_{kk}$ (inverse), … (4.3)

or $E_{lm}E_{ml} = E_{lm}(E_{lm})^{-1} = (E_{ml})^{-1}E_{ml} \equiv 1$, … (4.4)

iv. $E_{kl}E_{lm}E_{ml} = E_{kl}E_{lm}\ (E_{lm})^{-1} = E_{kl}$ (right identity), … (4.5)

and $E_{lk}E_{kl}E_{lm} = (E_{kl})^{-1}E_{kl}E_{lm} = E_{lm}$ (left identity), … (4.6)

v. $(E_{lm}E_{mn})^{-1} = (E_{mn})^{-1}(E_{lm})^{-1}$, … (4.7),

vi. $E_{jj} \equiv 1$ (identity equivalent to the scalar value). … (4.8)

(i) and (vi) are peculiar to the groupoid.

The operators in section 3 explicitly show the groupoid mentioned above. However, we have never calculated such matrices. Therefore, we need to check and verify that they really work.

Proof. At first, formula (i) is as follows:

a1). If $0 < l < m < n$ (mapping into higher dimensions),

$$x' = E_{lm}x = diag(\overbrace{1,1,\dots,1}^{l},\overbrace{D,D,\dots,D}^{m-l})(x_1,x_2,\dots,x_l,\overbrace{T,T,\dots,T}^{m-l})^T$$

$$= (x_1,x_2,\dots,x_l,\overbrace{DT,DT,\dots,T}^{m-l})^T$$

$$\therefore\ E_{mn}x' = diag(\overbrace{1,1,\dots,1}^{m},\overbrace{D,D,\dots,D}^{n-m})(x_1,x_2,\dots,x_l,\overbrace{DT,DT,\dots,DT}^{m-l}\overbrace{T,T,\dots,T}^{n-m})^T$$

$$= (x_1,x_2,\dots,x_l,\overbrace{DT,DT,\dots,T}^{n-l})^T$$

$$= diag(\overbrace{1,1,\dots,1}^{l},\overbrace{D,D,\dots,D}^{n-l})(x_1,x_2,\dots,x_l,\overbrace{T,T,\dots,T}^{n-l})^T$$

$$= E_{ln}x\ .$$

a2). If $0 < l < n < m$ (mapping into higher dimensions),

$$x' = E_{lm}x = diag(\overbrace{1,1,\dots,1}^{l},\overbrace{D,D,\dots,D}^{m-l})(x_1,x_2,\dots,x_l,\overbrace{T,T,\dots,T}^{m-l})^T$$

$$= (x_1,x_2,\dots,x_l,\overbrace{DT,DT,\dots,T}^{m-l})^T$$

$$\therefore\ E_{mn}x' = (E_{nm})^{-1}\,x'$$

$$= diag(\overbrace{1,1,\dots,1}^{n},\overbrace{D^{-1},D^{-1},\dots,D^{-1}}^{m-n})(x_1,x_2,\dots,x_l,\overbrace{DT,DT,\dots,DT}^{m-l}\,\overbrace{T,T,\dots,T}^{m-n})^T$$

$$= (x_1,x_2,\dots,x_l,\overbrace{DT,DT,\dots,DT}^{n-l}\,\overbrace{T,T,\dots,T}^{m-n})^T.$$

Remark. T is temporary. For example,

$$\begin{pmatrix}1 & 0 & 0\\ 0 & D^{-1} & 0\\ 0 & 0 & D^{-1}\end{pmatrix}\begin{pmatrix}a\\ DT\\ DT\end{pmatrix} = \begin{pmatrix}1 & 0 & 0\\ 0 & D^{-1} & 0\\ 0 & 0 & D^{-1}\end{pmatrix}\begin{pmatrix}1 & 0 & 0\\ 0 & D & 0\\ 0 & 0 & D\end{pmatrix}\begin{pmatrix}a\\ T\\ T\end{pmatrix} = \begin{pmatrix}1 & 0 & 0\\ 0 & 1 & 0\\ 0 & 0 & 1\end{pmatrix}\begin{pmatrix}a\\ T\\ T\end{pmatrix} = \begin{pmatrix}a\\ T\\ T\end{pmatrix} = a\,.$$

See also Eq. 3.1 and 3.5.

$$\therefore = (x_1, x_2, \dots, x_l, \overbrace{DT, DT, \dots, DT}^{n-l})^T$$

$$= diag(\overbrace{1,1, \dots, 1}^{l}, \overbrace{D, D, \dots, D}^{n-l})(x_1, x_2, \dots, x_l, \overbrace{T, T, \dots, T}^{n-l})^T$$

$$= E_{ln}x \ .$$

a3). If $0 < m < l < n$ (mapping into higher dimensions),

$$x' = E_{lm}x = (E_{ml})^{-1}\ x = diag(\overbrace{1,1, \dots, 1}^{m}, \overbrace{D^{-1}, D^{-1}, \dots, D^{-1}}^{l-m})(x_1, x_2, \dots, x_l)^T$$

$$= (x_1, x_2, \dots, x_m, D^{-1}x_{m+1}, D^{-1}x_{m+2}, \dots, D^{-1}x_l)^T.$$

$$\therefore\ E_{mn}x' = diag(\overbrace{1,1, \dots, 1}^{m}, \overbrace{D, \mathrm{D}, \dots, D}^{n-m})(x_1, x_2, \dots, x_m, \overbrace{D^{-1}x_{m+1}, D^{-1}x_{m+2}, \dots, D^{-1}x_l}^{l-m}, \overbrace{T, T, \dots, T}^{n-l})^T$$

$$= (x_1, x_2, \dots, x_l, \overbrace{DT, DT, \dots, DT}^{n-l})^T$$

$$= diag(\overbrace{1,1, \dots, 1}^{l}, \overbrace{D, D, \dots, D}^{n-l})(x_1, x_2, \dots, x_l, \overbrace{T, T, \dots, T}^{n-l})^T$$

$$= E_{ln}x \ .$$

b1). If $0 < n < m < l$ (mapping into lower dimensions),

$$x' = E_{lm}x = (E_{ml})^{-1}\ x = diag(\overbrace{1,1,\dots,1}^{m}, \overbrace{D^{-1}, D^{-1}, \dots, D^{-1}}^{l-m})(x_1, x_2, \dots, x_l)^T$$
$$= (x_1, x_2, \dots, x_m, D^{-1}x_{m+1}, D^{-1}x_{m+2}, \dots, D^{-1}x_l)^T.$$

$$\therefore\ E_{mn}x' = (E_{nm})^{-1}\ x'$$
$$= diag(\overbrace{1,1,\dots,1}^{n}, \overbrace{D^{-1}D^{-1}, \dots, D^{-1}}^{m-n}\ \overbrace{1,1,\dots,1}^{l-m})(x_1, x_2, \dots, x_m, D^{-1}x_{m+1}, D^{-1}x_{m+2}, \dots, D^{-1}x_l)^T$$
$$= (x_1, x_2, \dots, x_n, D^{-1}x_{n+1}, D^{-1}x_{n+2}, \dots, D^{-1}x_l)^T$$
$$= diag(\overbrace{1,1,\dots,1}^{n}, \overbrace{D^{-1}D^{-1}, \dots, D^{-1}}^{l-n})(x_1, x_2, \dots, x_l)^T$$
$$= (E_{nl})^{-1}x = E_{ln}x\ .$$

b2). If $0 < n < l < m$ (mapping into lower dimensions),

$$x' = E_{lm}x = diag(\overbrace{1,1,\dots,1}^{l}, \overbrace{D,D,\dots,D}^{m-l})(x_1, x_2, \dots, x_l, \overbrace{T,T,\dots,T}^{m-l})^T$$

$$= (x_1, x_2, \dots, x_l, \overbrace{DT, DT, \dots, DT}^{m-l})^T.$$

$$\therefore\ E_{mn}x' = (E_{nm})^{-1}\ x'$$

$$= diag(\overbrace{1,1,\dots,1}^{n}, \overbrace{D^{-1}D^{-1}, \dots, D^{-1}}^{m-n})\ (x_1, x_2, \dots, x_l, \overbrace{DT, DT, \dots, DT}^{m-l})^T$$

$$= (x_1, x_2, \dots, x_n, D^{-1}x_{n+1}, D^{-1}x_{n+2}, \dots, D^{-1}x_l, \overbrace{T, T, \dots, T}^{m-l})^T$$

$$= (x_1, x_2, \dots, x_n, D^{-1}x_{n+1}, D^{-1}x_{n+2}, \dots, D^{-1}x_l)^T$$

$$= diag(\overbrace{1,1,\dots,1}^{n}, \overbrace{D^{-1}D^{-1}, \dots, D^{-1}}^{l-n})(x_1, x_2, \dots, x_l)^T$$

$$= (E_{nl})^{-1}x = E_{ln}x\ .$$

b3). If $0 < m < n < l$ (mapping into lower dimensions),

$$x' = E_{lm}x = (E_{ml})^{-1}\, x = diag(\overbrace{1,1,\dots,1}^{m}, \overbrace{D^{-1}, D^{-1}, \dots, D^{-1}}^{l-m})(x_1, x_2, \dots, x_l)^T$$
$$= (x_1, x_2, \dots, x_m, D^{-1}x_{m+1}, D^{-1}x_{m+2}, \dots, D^{-1}x_l)^T.$$

$$\therefore\ E_{mn}x' = diag(\overbrace{1,1,\dots,1}^{m}, \overbrace{D, \mathrm{D}, \dots, D}^{n-m}\ \overbrace{1,1,\dots,1}^{n-l})(x_1, x_2, \dots, x_m, D^{-1}x_{m+1}, D^{-1}x_{m+2}, \dots, D^{-1}x_l)^T$$
$$= (x_1, x_2, \dots, x_n, D^{-1}x_{n+1}, D^{-1}x_{n+2}, \dots, D^{-1}x_l)^T$$
$$= diag(\overbrace{1,1,\dots,1}^{n}, \overbrace{D^{-1}D^{-1}, \dots, D^{-1}}^{l-n})(x_1, x_2, \dots, x_l)^T$$
$$= (E_{nl})^{-1}x = E_{ln}x\ .$$

Proof of (ii), the associative law is as follows: Since $(E_{kl}E_{lm})E_{mn} = E_{km}E_{mn} = E_{kn}$ and $E_{kl}(E_{lm}E_{mn}) = E_{kl}E_{ln} = E_{kn}$ from (i), $(E_{kl}E_{lm})E_{mn} = E_{kl}(E_{lm}E_{mn})$.

Proof of (iii) follows the rule of Eq. 3.6. Another proof is from the formula (i): $f(AA^{-1}) = I = f(A)f(A^{-1})$. $\therefore$ $f(A^{-1}) = f(A)^{-1}$, where $f(A^{-1}) = E_{ml}$ and $f(A)^{-1} = (E_{lm})^{-1}$.

Proof of (iv) is trivial from (ii).

Proof of (v), $E_{lm}E_{mn}(E_{lm}E_{mn})^{-1} = E_{lm}E_{mn}(E_{mn})^{-1}(E_{lm})^{-1} = E_{ll} \equiv 1$.

Finally, proof of (vi) is as follows. For $G = \{E_{lm}\}$, the scalar multiplication by 1 in field *k* holds as $s: 1 \times G = \mathrm{G} \times 1 \to G$. It is compatible with the matrix multiplications in *G* . Then,

$E_{mn} = I_m E_{mn} \equiv 1E_{mn} = E_{mn}1 \equiv E_{mn}I_n = E_{mn}$. Since it is 'mapping to itself' in the narrow sense of the word, it is equivalent to conventional unit matrices. □

Since this groupoid is homomorphism from (i), it can be considered as a representation of groupoid. Strictly speaking, it is automorphism. This will be proven later.

From Another Viewpoint

For E_{lm}, the binary operation is partially defined but not for any two elements arbitrarily taken from *G*. That is a groupoid. However, we must take note that group axioms do not claim that such a whole process ought to be done. To confirm that, let us try to give the following five conditions as group axioms:

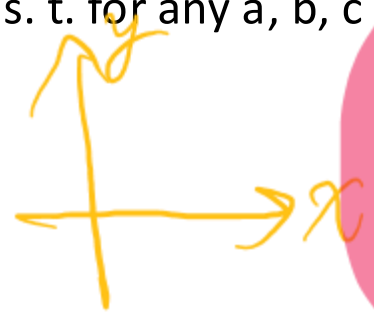
(1). We randomly take any two elements in a set *G*.

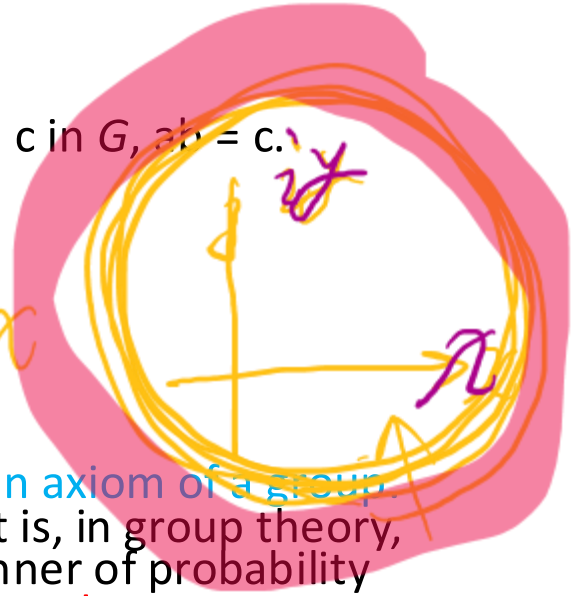
(2). For any two elements taken from *G*, the operation is closed in *G*, s. t. for any a, b, c in *G*, ab = c.

(3). For any *a, b, c* in *G*, $(ab)c = a(bc)$: associative law holds.

(4). There exists unique identity *e*.

(5). For each *a* in *G*, its inverse *b* exists s.t. $ab = ba = e$.





What we must pay attention to is whether the first condition should be included as an axiom of a group. If accepting that, we should introduce a concept of axioms in probability theory. That is, in group theory, we assume the whole event for any two elements arbitrarily taken from *G* in the manner of probability theory. Then, define binary operation as such for any elements taken from *G* at random. It means we should consider so-called 'sample space' or measure theory for probability in group theory. Group axioms naturally do not take such another axiomatic system in probability theory.

Remarks

There are **N** $\times$ **N** $\times$ **N** $\times$ **N** combinations of operations by E_{lm}. Even when the operations are valid there are **N** $\times$ **N** $\times$ **N** combinations. This obviously means that there are far more combinations than the ordinary algebraic operations in **N**, **Z**, **Q**, **R**, **C** and square matrices.

Nevertheless, **N** $\times$ **N** $\times$ **N** $\times$ **N** operations for all those elements create a 'sample space'. That is the 'axiom of probability theory', not an 'axiom of a group'.

Textbooks of algebra often write that the definition of a group is closed by $G \times G \rightarrow G$. That will be fine if the algebra you are dealing with is in **N**, **Z**, **Q**, **R**, **C** or square matrices. However, it violates the 'axioms of a group' and strengthens the conditions on their own.

If we interpret the axioms of a group correctly, it will be enough to say that $E_{ij}\ E_{jk}$ hold for any E_{lm} in G, which have already **N** $\times$ **N** $\times$ **N** combinations more than **N** $\times$ **N** combinations. This is the same as the logic of the associative law in linear algebra. That is, if $M_{jk} = A$, $M_{kl} = B$, and $M_{lm} = C$, then $M_{jk}M_{kl}M_{lm} = ABC = (AB)C = A(BC)$. In this case there are N $\times$ **N** $\times$ **N** $\times$ **N** combinations that are more than **N** $\times$ **N** $\times$ **N** combinations in the algebraic case ABC. Therefore, there is no problem.

However, considering E_{lm} to be a groupoid, more people will recognize and discuss it further. So we discuss it as a groupoid.

5. The Groupoid Representation

Claim. *Equation of the groupoid is $E_{lm}E_{mn} = E_{ln}$ automorphism.*

Proof. Firstly, the automorphism is proven as follows. Let $f(A)$ be E_{lm}, $f(B)$ be E_{mn}.

a1). If $0 < l < m < n$ (mapping into higher dimensions),

For $f(A)(B)$,

$$(f(A))(x) = x' = E_{lm}x = diag(\overbrace{1,1,\dots,1}^{l}, \overbrace{D,D,\dots,D}^{m-l})(x_1, x_2, \dots, x_l, \overbrace{T,T,\dots,T}^{m-l})^T$$

$$= (x_1, x_2, \dots, x_l, \overbrace{DT, DT, \dots, T}^{m-l})^T.$$

Then $\big(f(B)\big)(x') = E_{mn}x' = diag(\overbrace{1,1,\dots,1}^{m}, \overbrace{D,D,\dots,D}^{n-m})(x_1, x_2, \dots, x_l, \overbrace{DT, DT, \dots, DT}^{m-l}\overbrace{T,T,\dots,T}^{n-m})^T$

$$= (x_1, x_2, \dots, x_l, \overbrace{DT, DT, \dots, T}^{n-l})^T$$

$$= diag(\overbrace{1,1,\dots,1}^{l}, \overbrace{D,D,\dots,D}^{n-l})(x_1, x_2, \dots, x_l, \overbrace{T,T,\dots,T}^{n-l})^T$$

$$= E_{ln}x\ .$$

For $f(AB)$,

$$\big(f(AB)\big)(x) = E_{lm}E_{mn}x$$

$$= diag(\overbrace{1,1,\dots,1}^{l},\overbrace{D,D,\dots,D}^{m-l}\overbrace{1,1,\dots,1}^{n-m})diag(\overbrace{1,1,\dots,1}^{m},\overbrace{D,D,\dots,D}^{n-m})(x_1,x_2,\dots,x_l,\overbrace{T,T,\dots,T}^{n-l})^T$$

$$= (x_1,x_2,\dots,x_l,\overbrace{DT,DT,\dots,T}^{n-l})^T$$

$$= diag(\overbrace{1,1,\dots,1}^{l},\overbrace{D,D,\dots,D}^{n-l})(x_1,x_2,\dots,x_l,\overbrace{T,T,\dots,T}^{n-l})^T$$

$$= E_{ln}x\ .$$

$$\therefore f(AB) = f(A)(B).$$

a2). If $0 < l < n < m$ (mapping into higher dimensions),

For $f(A)f(B)$,

$$(f(A))(x) = x' = E_{lm}x$$

$$= diag(\overbrace{1,1,\dots,1}^{l},\overbrace{D,D,\dots,D}^{m-l})(x_1,x_2,\dots,x_l,\overbrace{T,T,\dots,T}^{m-l})^T$$

$$= (x_1,x_2,\dots,x_l,\overbrace{DT,DT,\dots,T}^{m-l})^T$$

Then $\big(f(B)\big)(x') = E_{mn}x' = (E_{nm})^{-1}\,x'$

$$= diag(\overbrace{1,1,\dots,1}^{n},\overbrace{D^{-1},D^{-1},\dots,D^{-1}}^{m-n})(x_1,x_2,\dots,x_l,\overbrace{DT,DT,\dots,DT}^{m-l})^T$$

$$= (x_1,x_2,\dots,x_l,\overbrace{DT,DT,\dots,DT}^{n-l}\,\overbrace{T,T,\dots,T}^{m-n})^T$$

$$= (x_1,x_2,\dots,x_l,\overbrace{DT,DT,\dots,DT}^{n-l})^T$$

$$= diag(\overbrace{1,1,\dots,1}^{l},\overbrace{D,D,\dots,D}^{n-l})(x_1,x_2,\dots,x_l,\overbrace{T,T,\dots,T}^{n-l})^T$$

$$= E_{ln}x\ .$$

For $f(AB)$,

$$\big(f(AB)\big)(x) = (E_{lm}E_{mn})x$$

$$= diag(\overbrace{1,1,\dots,1}^{l},\overbrace{D,D,\dots,D}^{m-l})diag(\overbrace{1,1,\dots,1}^{n},\overbrace{D^{-1},D^{-1},\dots,D^{-1}}^{m-n})(x_1,x_2,\dots,x_l,\overbrace{T,T,\dots,T}^{m-l})^T$$

$$= (x_1,x_2,\dots,x_l,\overbrace{DT,DT,\dots,DT}^{n-l}\overbrace{T,T,\dots,T}^{m-n})^T$$

$$= (x_1,x_2,\dots,x_l,\overbrace{DT,DT,\dots,DT}^{n-l})^T$$

$$= diag(\overbrace{1,1,\dots,1}^{l},\overbrace{D,D,\dots,D}^{n-l})(x_1,x_2,\dots,x_l,\overbrace{T,T,\dots,T}^{n-l})^T$$

$$= E_{ln}x\ .$$

$$\therefore f(AB) = f(A)(B).$$

a3). If $0 < m < l < n$ (mapping into higher dimensions),

For $f(A)f(B)$,

$$(f(A))(x) = x' = E_{lm}x = (E_{ml})^{-1}\ x$$

$$= diag(\overbrace{1,1,\dots,1}^{m}, \overbrace{D^{-1}, D^{-1}, \dots, D^{-1}}^{l-m})(x_1, x_2, \dots, x_l)^T$$

$$= (x_1, x_2, \dots, x_m, D^{-1}x_{m+1}, D^{-1}x_{m+2}, \dots, D^{-1}x_l)^T.$$

Then, $\big(f(B)\big)(x') = E_{mn}x'$

$$= diag(\overbrace{1,1,\dots,1}^{m}, \overbrace{D, \mathrm{D}, \dots, D}^{n-m})(x_1, x_2, \dots, x_m, \overbrace{D^{-1}x_{m+1}, D^{-1}x_{m+2}, \dots, D^{-1}x_l}^{l-m}, \overbrace{T, T, \dots, T}^{n-l})^T$$

$$= (x_1, x_2, \dots, x_l, \overbrace{DT, DT, \dots, DT}^{n-l})^T$$

$$= diag(\overbrace{1,1,\dots,1}^{l}, \overbrace{D, D, \dots, D}^{n-l})(x_1, x_2, \dots, x_l, \overbrace{T, T, \dots, T}^{n-l})^T$$

$$= E_{ln}x\ .$$

For $f(AB)$,

$$\big(f(AB)\big)(x) = (E_{lm}E_{mn})x$$

$$= diag(\overbrace{1,1,\dots,1}^{m}, \overbrace{D^{-1}, D^{-1}, \dots, D^{-1}}^{l-m} \overbrace{1,1,\dots,1}^{n-l}) diag(\overbrace{1,1,\dots,1}^{m}, \overbrace{D, \mathrm{D}, \dots, D}^{n-m})(x_1, x_2, \dots, x_l \overbrace{T, T, \dots, T}^{n-l})^T$$

$$= (x_1, x_2, \dots, x_l, \overbrace{DT, DT, \dots, DT}^{n-l})^T$$

$$= diag(\overbrace{1,1,\dots,1}^{l}, \overbrace{D, \mathrm{D}, \dots, D}^{n-l})(x_1, x_2, \dots, x_l \overbrace{T, T, \dots, T}^{n-l})^T$$

$$= E_{ln}x\ .$$

$$\therefore f(AB) = f(A)(B).$$

b1). If $0 < n < m < l$ (mapping into lower dimensions),

For $f(A)f(B)$,

$$(f(A))(x) = x'z = E_{lm}x = (E_{ml})^{-1}\ x = diag(\overbrace{1,1,\dots,1}^{m}, \overbrace{D^{-1}, D^{-1}, \dots, D^{-1}}^{l-m})(x_1, x_2, \dots, x_l)^T$$
$$= (x_1, x_2, \dots, x_m, D^{-1}x_{m+1}, D^{-1}x_{m+2}, \dots, D^{-1}x_l)^T.$$

Then,

$$\big(f(B)\big)(x) = E_{mn}x' = (E_{nm})^{-1}\ x'$$
$$= diag(\overbrace{1,1,\dots,1}^{n}, \overbrace{D^{-1}D^{-1}, \dots, D^{-1}}^{m-n} \overbrace{1,1,\dots,1}^{l-m})(x_1, x_2, \dots, x_m, D^{-1}x_{m+1}, D^{-1}x_{m+2}, \dots, D^{-1}x_l)^T$$
$$= (x_1, x_2, \dots, x_n, D^{-1}x_{n+1}, D^{-1}x_{n+2}, \dots, D^{-1}x_l)^T$$
$$= diag(\overbrace{1,1,\dots,1}^{n}, \overbrace{D^{-1}D^{-1}, \dots, D^{-1}}^{l-n})(x_1, x_2, \dots, x_l)^T$$
$$= (E_{nl})^{-1}x = E_{ln}x\ .$$

For $f(AB)$,

$$\big(f(AB)\big)(x) = (E_{lm}E_{mn})x$$

$$= diag(\overbrace{1,1,\dots,1}^{m},\overbrace{D^{-1},D^{-1},\dots,D^{-1}}^{l-m})\, diag(\overbrace{1,1,\dots,1}^{n},\overbrace{D^{-1}D^{-1},\dots,D^{-1}}^{m-n}\overbrace{1,1,\dots,1}^{l-m})(x_1,x_2,\dots,x_l)^T$$

$$= (x_1,x_2,\dots,x_n,D^{-1}x_{n+1},D^{-1}x_{n+2},\dots,D^{-1}x_l)^T$$

$$= diag(\overbrace{1,1,\dots,1}^{n},\overbrace{D^{-1}D^{-1},\dots,D^{-1}}^{l-n})(x_1,x_2,\dots,x_l)^T$$

$$= (E_{nl})^{-1}x = E_{ln}x\ .$$

$$\therefore f(AB) = f(A)(B).$$

b2). If $0 < n < l < m$ (mapping into lower dimensions),

For $f(A)f(B)$,

$$(f(A))(x) = x' = E_{lm}x = diag(\overbrace{1,1,\dots,1}^{l}, \overbrace{D, \mathrm{D}, \dots, D}^{m-l})(x_1, x_2, \dots, x_l, \overbrace{T, T, \dots, T}^{m-l})^T$$

$$= (x_1, x_2, \dots, x_l, \overbrace{DT, DT, \dots, DT}^{m-l})^T.$$

Then,

$$(f(B))(x') = E_{mn}x' = (E_{nm})^{-1}\, x'$$

$$= diag(\overbrace{1,1,\dots,1}^{n}, \overbrace{D^{-1}D^{-1}, \dots, D^{-1}}^{m-n})(x_1, x_2, \dots, x_l, \overbrace{DT, DT, \dots, DT}^{m-l})^T$$

$$= (x_1, x_2, \dots, x_n, D^{-1}x_{n+1}, D^{-1}x_{n+2}, \dots, D^{-1}x_l, \overbrace{T, T, \dots, T}^{m-l})^T$$

$$= (x_1, x_2, \dots, x_n, D^{-1}x_{n+1}, D^{-1}x_{n+2}, \dots, D^{-1}x_l)^T$$

$$= diag(\overbrace{1,1,\dots,1}^{n}, \overbrace{D^{-1}D^{-1}, \dots, D^{-1}}^{l-n})(x_1, x_2, \dots, x_l)^T$$

$$= (E_{nl})^{-1}x = E_{ln}x\ .$$

For $f(AB)$,

$$\big(f(AB)\big)(x) = (E_{lm}E_{mn})x$$

$$= diag(\overbrace{1,1,\dots,1}^{l}, \overbrace{D, \mathrm{D}, \dots, D}^{m-l}) diag(\overbrace{1,1,\dots,1}^{n}, \overbrace{D^{-1}D^{-1}, \dots, D^{-1}}^{m-n})(x_1, x_2, \dots, x_l, \overbrace{T, T, \dots, T}^{m-l})^T$$

$$= (x_1, x_2, \dots, x_n, D^{-1}x_{n+1}, D^{-1}x_{n+2}, \dots, D^{-1}x_l, \overbrace{T, T, \dots, T}^{m-l})^T$$

$$= (x_1, x_2, \dots, x_n, D^{-1}x_{n+1}, D^{-1}x_{n+2}, \dots, D^{-1}x_l)^T$$

$$= diag(\overbrace{1,1,\dots,1}^{n}, \overbrace{D^{-1}D^{-1}, \dots, D^{-1}}^{l-n})(x_1, x_2, \dots, x_l)^T$$

$$= (E_{nl})^{-1}x = E_{ln}x \ .$$

$$\therefore f(AB) = f(A)(B).$$

b3). If $0 < m < n < l$ (mapping into lower dimensions),

For $f(A)f(B)$,

$$(f(A))(x) = x' = E_{lm}x = (E_{ml})^{-1}\ x$$

$$= diag(\overbrace{1,1,\dots,1}^{m}, \overbrace{D^{-1}, D^{-1}, \dots, D^{-1}}^{l-m})(x_1, x_2, \dots, x_l)^T$$

$$= (x_1, x_2, \dots, x_m, D^{-1}x_{m+1}, D^{-1}x_{m+2}, \dots, D^{-1}x_l)^T.$$

Then,

$$\big(f(B)\big)(x') = E_{mn}x'$$

$$= diag(\overbrace{1,1,\dots,1}^{m}, \overbrace{D, \mathrm{D}, \dots, D}^{n-m}\ \overbrace{1,1,\dots,1}^{l-n})(x_1, x_2, \dots, x_m, D^{-1}x_{m+1}, D^{-1}x_{m+2}, \dots, D^{-1}x_l)^T$$

$$= (x_1, x_2, \dots, x_n, D^{-1}x_{n+1}, D^{-1}x_{n+2}, \dots, D^{-1}x_l)^T$$

$$= diag(\overbrace{1,1,\dots,1}^{n}, \overbrace{D^{-1}D^{-1}, \dots, D^{-1}}^{l-n})(x_1, x_2, \dots, x_l)^T$$

$$= (E_{nl})^{-1}x = E_{ln}x\ .$$

For $f(AB)$,

$$(f(AB))(x) = (E_{lm}E_{mn})x$$

$$= diag(\overbrace{1,1,\dots,1}^{m}, \overbrace{D, \mathrm{D}, \dots, D}^{l-m})diag(\overbrace{1,1,\dots,1}^{m}, \overbrace{D, \mathrm{D}, \dots, D}^{n-m}\overbrace{1,1,\dots,1}^{l-n})(x_1, x_2, \dots, x_l)^T$$

$$= (x_1, x_2, \dots, x_n, D^{-1}x_{n+1}, D^{-1}x_{n+2}, \dots, D^{-1}x_l)^T$$

$$= diag(\overbrace{1,1,\dots,1}^{n}, \overbrace{D^{-1}D^{-1}, \dots, D^{-1}}^{l-n})(x_1, x_2, \dots, x_l)^T$$

$$= (E_{nl})^{-1}x = E_{ln}x\ .$$

$$\therefore f(AB) = f(A)(B).$$

6. The Finite Simple Groupoid

A modelling of the Transformation Groupoid

We discuss a movement as follows. There is a point on the whole number line. We move it at an integer m to another one n on that line. We denote it $_iT_j$. We think of the whole elements $G = \{_iT_j | i, j \in \mathrm{Z}\}$. Then, we discuss the partial operation $_iT_k$= $_iT_{jj}T_k$. It expresses repetition of the movements. *G* makes a groupoid. This is a modelling of the groupoid on the line as mentioned in former chapters. In this chapter, we think of the subgroupoids.

Introduction

Thinking of a point on the whole number line, we move the point at the integer m to another one n on the number line, as shown in Fig. 16. We denote the movement ${}_mT_n$.

mTn

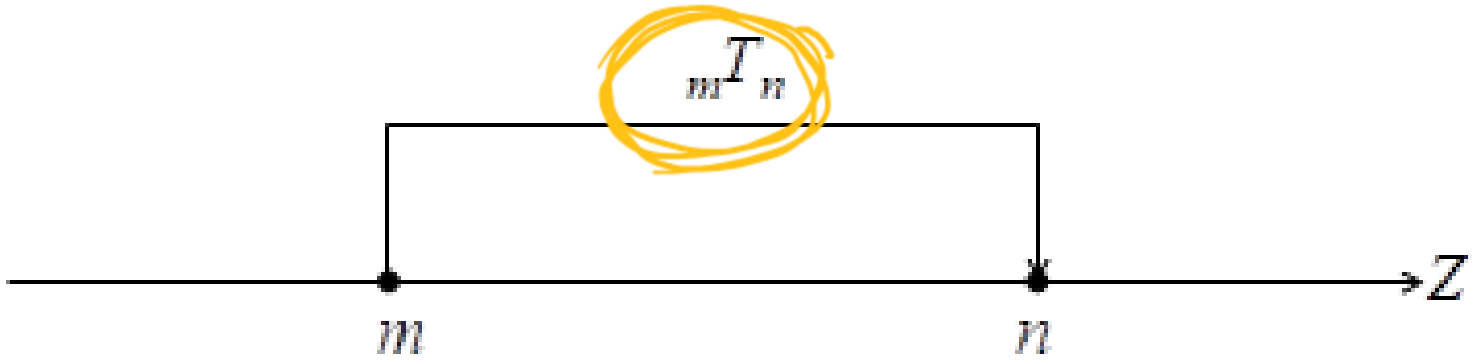
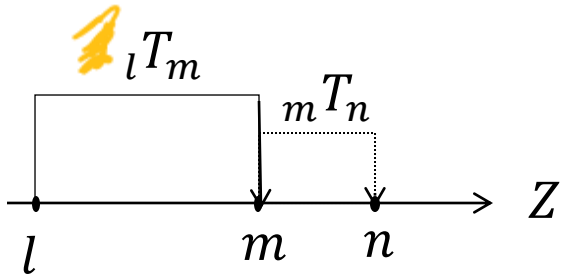
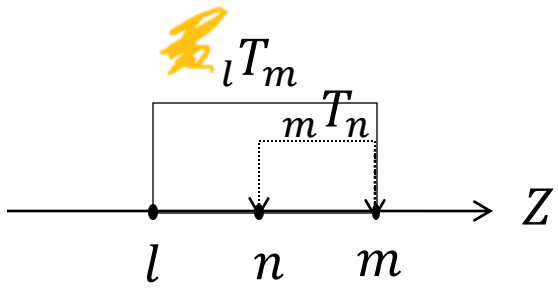


Fig. 16

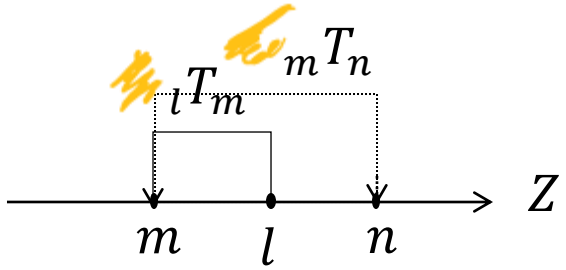
We think of all the elements $G = \{{}_mT_n | m, n \in \mathrm{Z}\}$. Now, let us think of the partial function for the repetition of the movement ${}_lT_n = {}_l\,T_{mm}T_n$. It means a series of movements of a point at the integer l to m, then m to n on the whole number line. There are 6 types of procedures for ${}_lT_n = {}_l\,T_{mm}T_n$ as shown in Fig. 17.



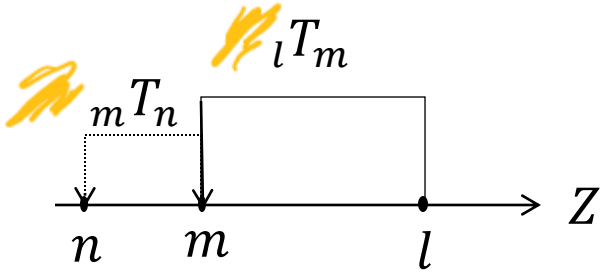
$_lT_n = {}_l T_m {}_mT_n \ (l \le m \le n)$



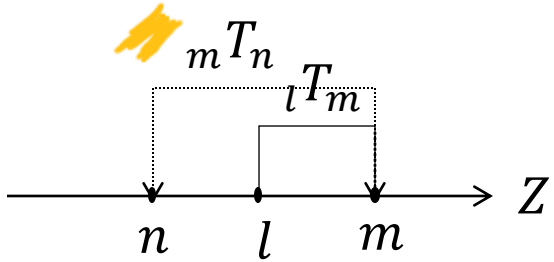
$_lT_n = {}_l T_m {}_mT_n \ (l \le n \le m)$



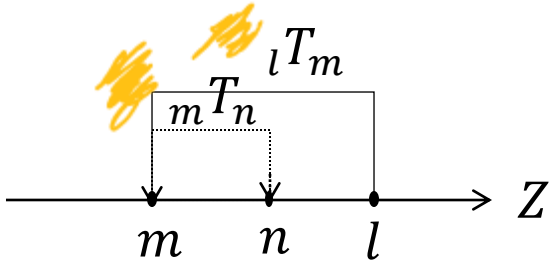
$_lT_n = {}_l T_m {}_mT_n \ (m \le l \le n)$



$_lT_n = {}_l T_m {}_mT_n \ (n \le m \le l)$



$_lT_n = {}_l T_m {}_mT_n \ (n \le l \le m)$



$_lT_n = {}_l T_m {}_mT_n \ (m \le n \le l)$

Fig. 17

Theorem 6. *Laws under the rule of partial operations in G are as follows:*

$\times$ $N \times N$

I. ${}_iT_{jj}T_k = {}_i\,T_k$, … (6.1)

II. $({}_iT_{jj}T_k)_kT_l = {}_i\,T_j({}_jT_{kk}T_l)$, … (6.2)

III. $({}_iT_j)^{-1} = {}_j\,T_i$, … (6.3)

IV. ${}_iT_{jj}T_{kk}T_j = {}_i\,T_{jj}T_k({}_jT_k)^{-1} = {}_i\,T_j$ (right identity), … (6.4)

and ${}_jT_{ii}T_{jj}T_k = ({}_iT_j)^{-1}\,{}_i\,T_{jj}T_k = {}_j\,T_k$ (left identity), … (6.5)

V. ${}_iT_{jj}T_i = {}_i\,T_i \equiv 1 \equiv {}_j\,T_{ii}T_j = {}_j\,T_j$. … (6.6)

Proof.

(I) is trivial by the definition (see also chap. 3).

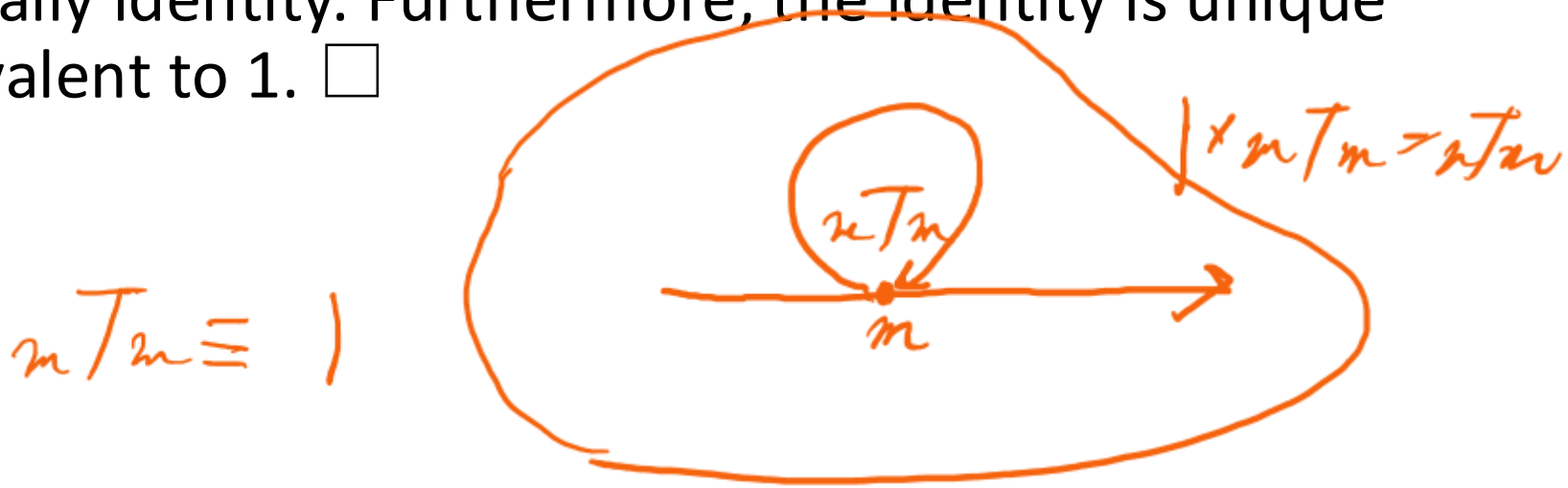
The proof of (II) is as follows. From (I),

$({}_iT_{jj}T_k)_kT_l = {}_i\,T_{kk}T_l = {}_i\,T_l$, and ${}_iT_j({}_jT_{kk}T_l) = {}_i\,T_{jj}T_l = {}_i\,T_l$.

(III) is clear under the operation on the whole number line.

(IV) is trivial from (I).

The proof of (V) is as follows. Since ${}_mT_m$ is 'mapping to itself' on the number line, it is naturally identity. Furthermore, the identity is unique because they are equivalent to 1. □



The Subgroupoids

Let us think of the subgroupoids. They mainly consist of two subgroupoids:

$H = \{{}_{2m-1}T_{2n-1} | m, n \in \mathrm{Z}\}$, … (6.7)

$H' = \{{}_{2m}T_{2n} | m, n \in \mathrm{Z}\}$. … (6.8)

The rest $G - (H \cup H') = \{{}_{2i-1}T_{2j}, {}_{2k}T_{2l-1} | i, j, k, l \in \mathrm{Z}\}$ … (6.9)

are not subgroupoids but subsets.

***Remark*.** If $\{{}_{2m-1}T_{2n}\}\{{}_{2n}T_{2m-1}\} = \{{}_{2m-1}T_{2m-1}\} = 1$, then they are not in $G - (H \cup H')$.

We actually need to pay attention to such an operation. If we take the elements like ${}_4T_{33}T_2 = {}_4T_2$, it is not in $G - (H \cup H')$ but *H'* . These operations are equivalent to operations in *G* . Eventually, *H*, *H'*, and $G - (H \cup H')$ are disjoint. Therefore, the direct sum decomposition is:

$$G = \{{}_{2m-1}T_{2n-1}\} \sqcup \{{}_{2m}T_{2n}\} \sqcup \{{}_{2i-1}T_{2j}\} \sqcup \{{}_{2k}T_{2l-1}\} \sqcup \{{}_{u}T_{u}\}. \quad \ldots (6.10)$$

Proposition 7.1. *H and H' are not normal subgroupoids of G.* ${}_3T_2$ $g = {}_1T_3$

***Proof*.** Let us assume that *H* is a normal subgroupoid. Let *g*, for example, be any element in $\{{}_{2m-1}T_{2n}\}$. Operating *gH* is impossible unless any element in *H* is ${}_{2n}T_{2l-1}$ or ${}_{2n}T_{2l}$ under the rule of the partial function. Therefore, *gH* cannot be left coset. $\therefore gH \notin G$ in this case. Operating *Hg*, because of $H = \{{}_{2k-1}T_{2m-1}\}$, it holds and the right coset $Hg = \{{}_{2k-1}T_{2n}\}$. g ${}_1T_3\,{}_3T_2 = {}_1T_2$

$\therefore gH \neq Hg$. $gH \neq Hg$ ${}_3T_2\,{}_1T_3$ -

Likewise, let us assume that *H'* is a normal subgroupoid. Let, for example, $\forall g' \in \{{}_{2m}T_{2n-1}\}$. Operating *g'H'*, it cannot be operated unless any elements in *H'* are $\{{}_{2n-1}T_{2l}\}$ or $\{{}_{2n-1}T_{2l-1}\}$. They are not in *H'* either. $\therefore g'H' \notin G$. Operating *H'g'*, *H'g'* holds since any elements in *H'* are $\{{}_{2k}T_{2m}\}$. Then, $H'g' = \{{}_{2k}T_{2n-1}\}$.

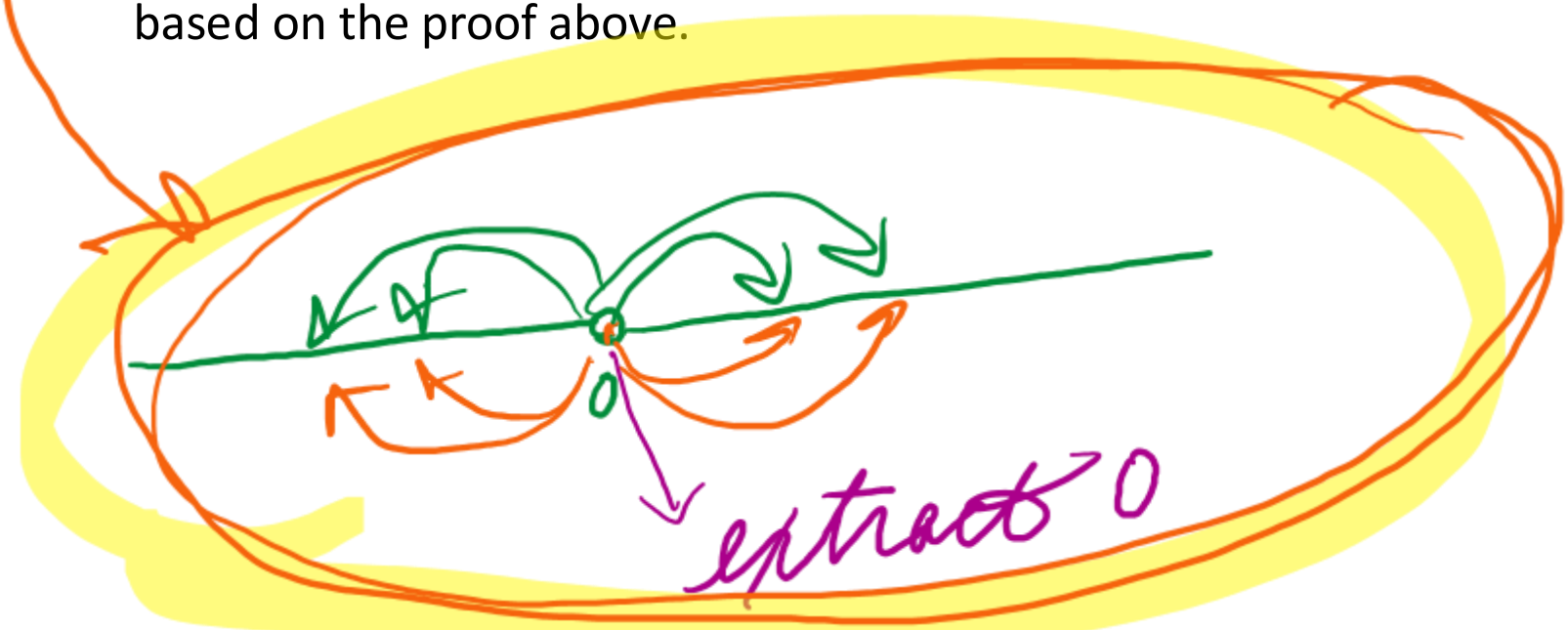
$\therefore g'H' \neq H'g'$. $\square$

As another way of creating subgroupoids from the groupoid, let us think of $H'' = \{{}_mT_n | k < m < l, k < n < l, k, l, m, n \in \mathrm{Z}\}$. However, none of them are normal subgroupoids other than the identity: For H'', only if $m = n$(: $H''\{{}_lT_l\} \equiv 1$), then $gH'' = H''g$. It is trivial ($g_{kl}1 = 1g_{kl} = g_{kl}$).

Proposition 7.2. The maximal proper subgroupoid *H '''* is a set of extracting elements whose either index number is fixed and all the inverses from *G:* any element should have inverse also in subgroupoid by two-step 'subgroupoid' test. For example $H''' = G - (\{_mT_0\} \cup \{_0T_m\})$. It graphically means only the point '0' is skipped in a series of transformation on the number line. Likewise, it is also not the normal subgroupoid.

Proof. $_mT_0H'''$ does not hold but H'''_mT_0 does. $\therefore$ $_m\ T_0H''' \neq H'''_mT_0$. $\square$

From this proof, it is clear that any other smaller subgroupoids extracting any other subsets or families of sets will not be normal. The simplest example is a case of $G = \{_mT_n | m, n = 0,1,2\}$ and the smallest subgroupoid (except for the identity as subgroupoid) is $\{_mT_n | m, n = 0,1\}$, $\{_mT_n | m, n = 1,2\}$, and $\{_mT_n | m, n = 0,2\}$. It is trivial that none of these subgroupoids are normal based on the proof above.



Contrarily, what will happen if we sum up those extracted families of sets? What we get interested in will be if such a summed-up subset makes a subgroupoid of G. To confirm it, let us think of such a total subset s_k as follows.

$$\boldsymbol{s_k} = (\{{}_{\boldsymbol{m}}\boldsymbol{T}_{-|\boldsymbol{m}|}\} \cup \{{}_{-|\boldsymbol{m}|}\boldsymbol{T}_{\boldsymbol{m}}\}) \cup \cdots \cup (\{{}_{\boldsymbol{m}}\boldsymbol{T}_{-1}\} \cup \{{}_{-1}\boldsymbol{T}_{\boldsymbol{m}}\}) \cup (\{{}_{\boldsymbol{m}}\boldsymbol{T}_{0}\} \cup \{{}_{0}\boldsymbol{T}_{\boldsymbol{m}}\}) \cup (\{{}_{\boldsymbol{m}}\boldsymbol{T}_{1}\} \cup \{{}_{1}\boldsymbol{T}_{\boldsymbol{m}}\}) \cup \cdots \cup (\{{}_{\boldsymbol{m}}\boldsymbol{T}_{\boldsymbol{k}}\} \cup \{{}_{\boldsymbol{k}}\boldsymbol{T}_{\boldsymbol{m}}\} \quad (\mathbf{5.11})$$

where $k \leq m$. If $k = m$, $s_k = s_m = G$.

If corresponding this model on the line to the groupoid that we have discussed, then $0 \leq k \leq m$.

$$\therefore \boldsymbol{s_k} = (\{{}_{\boldsymbol{m}}\boldsymbol{T}_{\boldsymbol{0}}\} \cup \{{}_{\boldsymbol{0}}\boldsymbol{T}_{\boldsymbol{m}}\}) \cup (\{{}_{\boldsymbol{m}}\boldsymbol{T}_{\boldsymbol{1}}\} \cup \{{}_{\boldsymbol{1}}\boldsymbol{T}_{\boldsymbol{m}}\}) \cup \cdots \cup (\{{}_{\boldsymbol{m}}\boldsymbol{T}_{\boldsymbol{k}}\} \cup \{{}_{\boldsymbol{k}}\boldsymbol{T}_{\boldsymbol{m}}\}. \quad \ldots (6.12)$$

If thinking of the partial operation ${}_{k+1}T_{kk}T_{k+2}$, then

${}_{k+1}T_{kk}T_{k+2} = {}_{k+1}T_{k+2} \notin s_k$. Besides, if ${}_{m}T_{m-1m-1}T_{m}$, then ${}_{m}T_{m-1m-1}T_{m} = {}_{m}T_{m} \notin s_{m-1}$.

Thus, any such total subsets do not make the subgroupoid. If adding ${}_{m}T_{m}$ on to s_{m-1} , $s_{m-1} \sqcup {}_{m}T_{m}$ results in G . Verifying them on the number line will be clearer.

For Your Information

We resulted in proving that $_lT_m$ is a finite simple ‘group’, not only a finite simple groupoid (see also slide 46 to 47). That is so-called ‘Conway’s nightmare’ among group theorists researching the classification of finite simple groups. Most of these theorists believe that any new finite simple groups do not exist. However, some are apprehensive about the possible existence of such groups. John Horton Conway of ‘Conway’s nightmare’ is known as one of the contributors of finite simple groups.

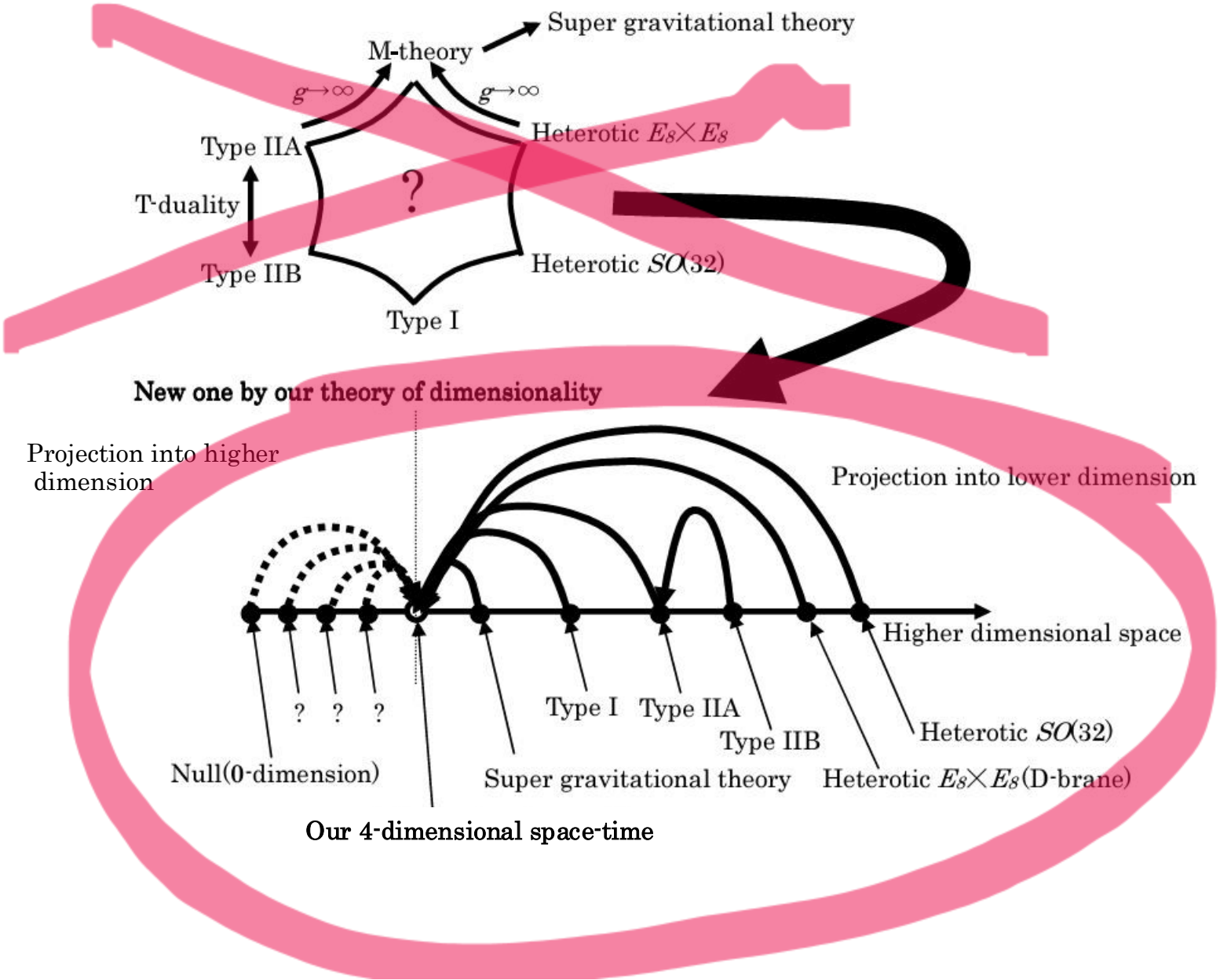
7. The Invariant and Symmetry — Towards Noether's Theorem

- **Invariant:** in theoretical physics, an invariant means a physical system unchanged under mathematical operation. This is also called symmetry.
- **Noether's theorem:** this states that every differential symmetry of the action of a physical system with conservative forces has a corresponding law of conservation.
- Noether's theorem actually holds not only differential symmetries but also discrete symmetries. Parity and selection rule in quantum theory are such examples.

What is the Invariant in the Groupoid that we have discussed?

- The invariant is the conservation of the degree of freedom. Wherever a point is projected, its degrees of freedom are conserved.
- It suggests that if higher dimensional physics were described by the groupoid, we might find an unknown physical law of conservation.

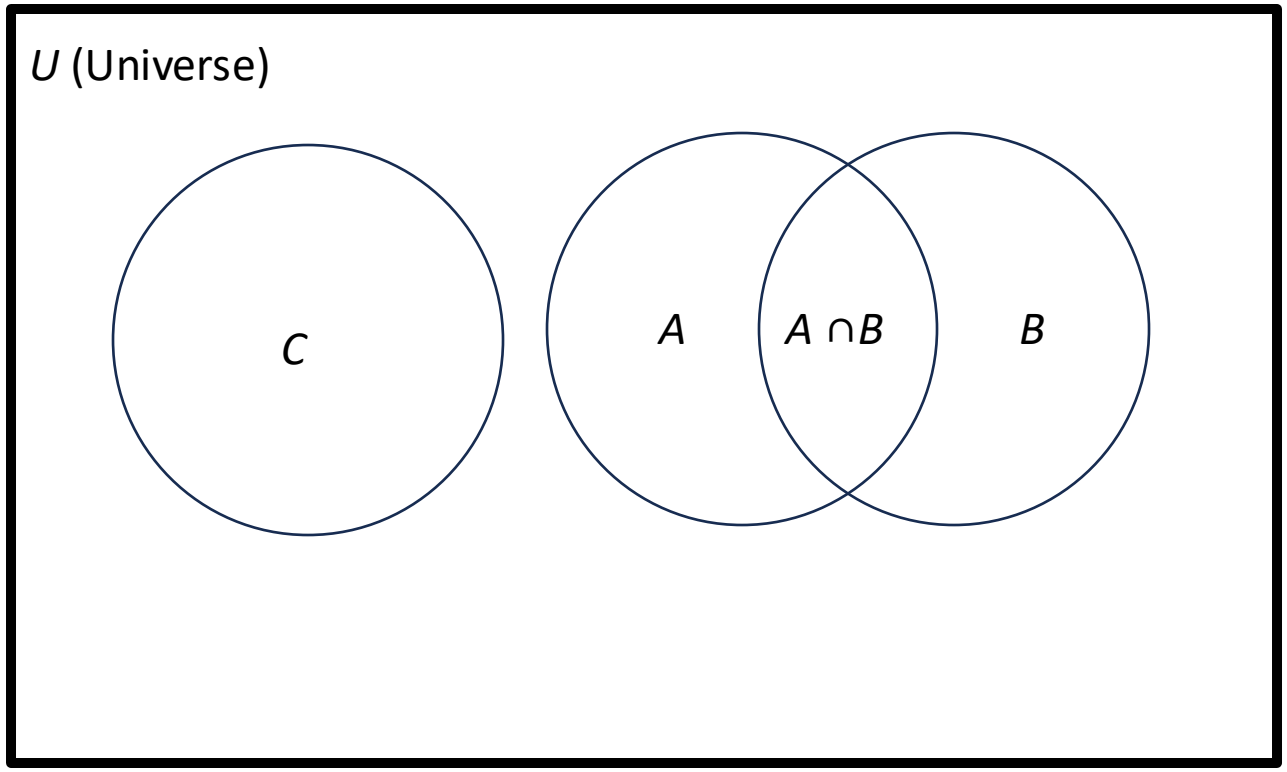
What is a Preferable Unification of String Theories?



- The upper model is by M-Theory.
- The lower one is by our theory of dimensionality.

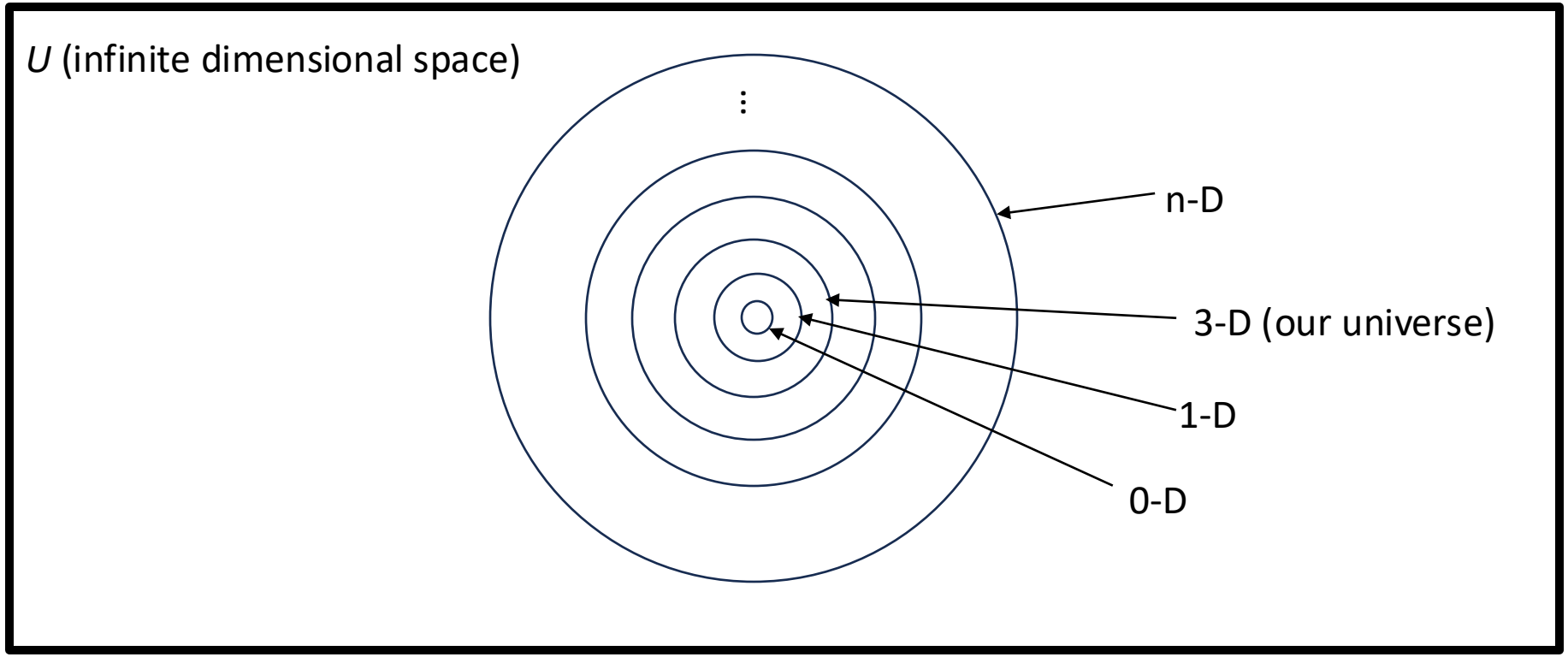
8. The Set Theoretical Approach

Remark. Topological space, that is the concept of distance, is NOT defined in set theory. There is no concept of neighbourhood. *A, B* and *C* are not Hausdorff space.



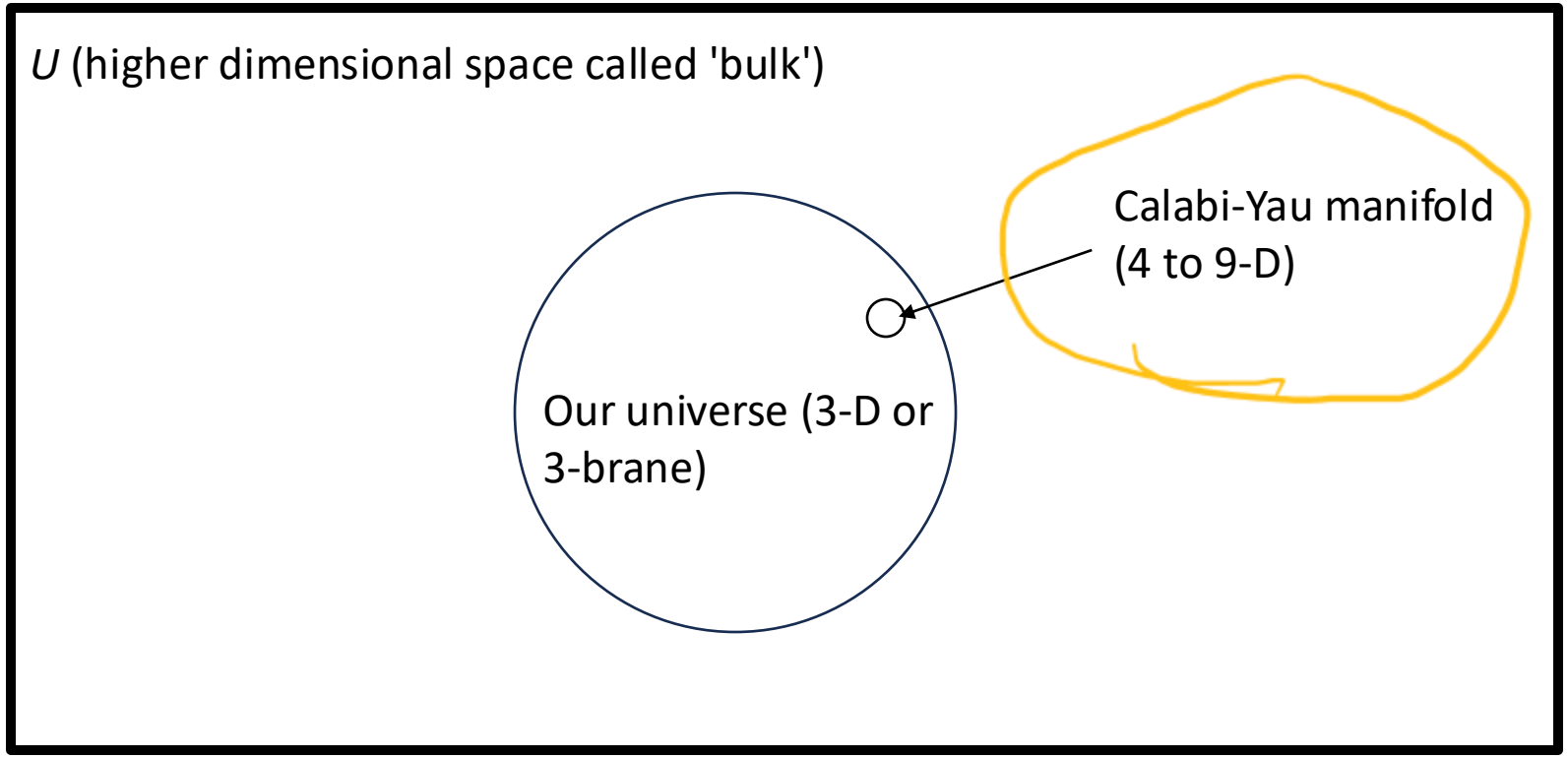
Empirical Thought of Dimensionality

We have never detected higher dimensional space, much less infinite dimensional space.



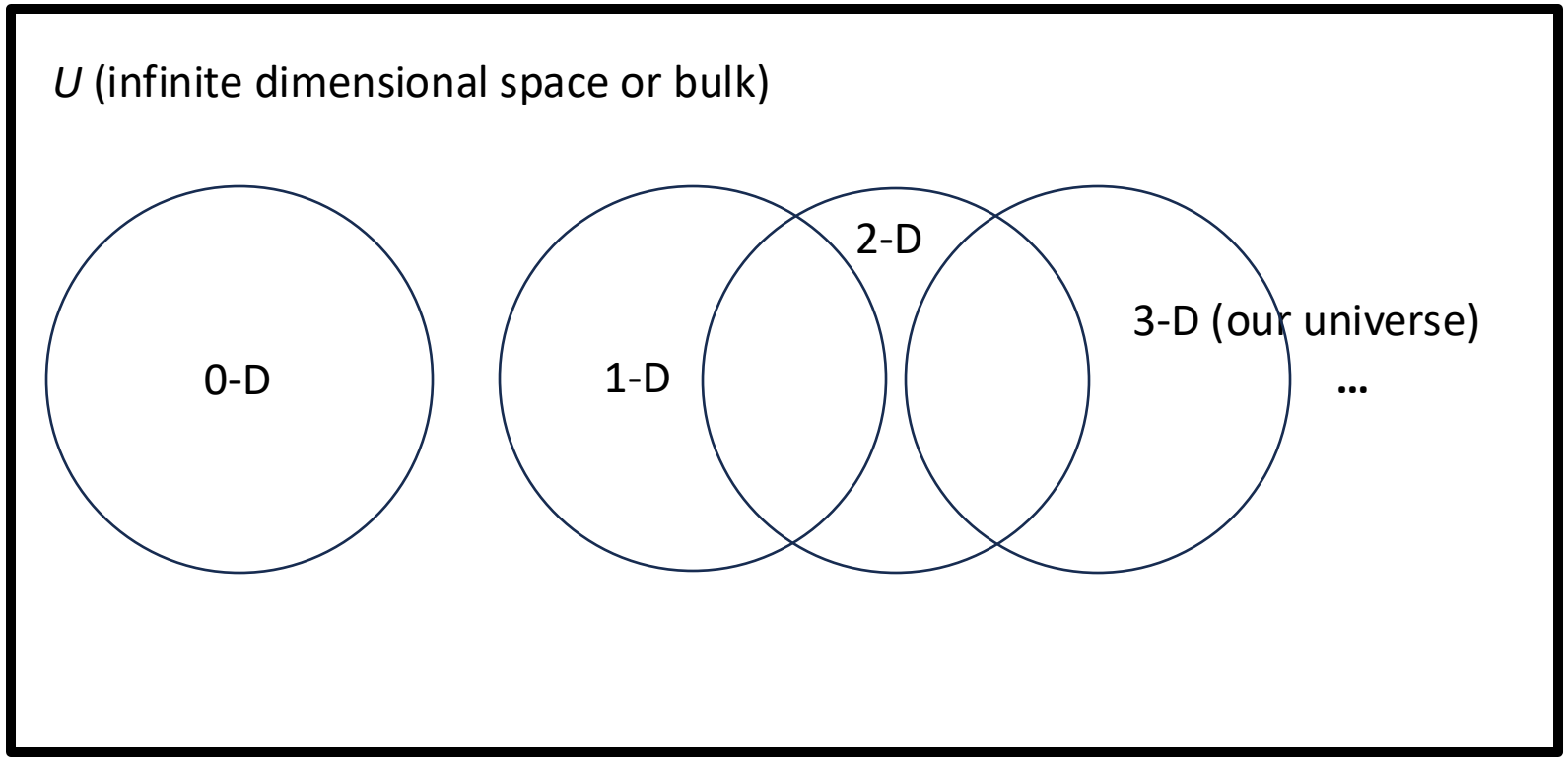
M-theory

M-theory says that our universe (3-brane) drifts on the bulk. However, we have neither detected Calabi-Yau manifold nor bulk.



Still Empirical but Rational Thought of Dimensionality

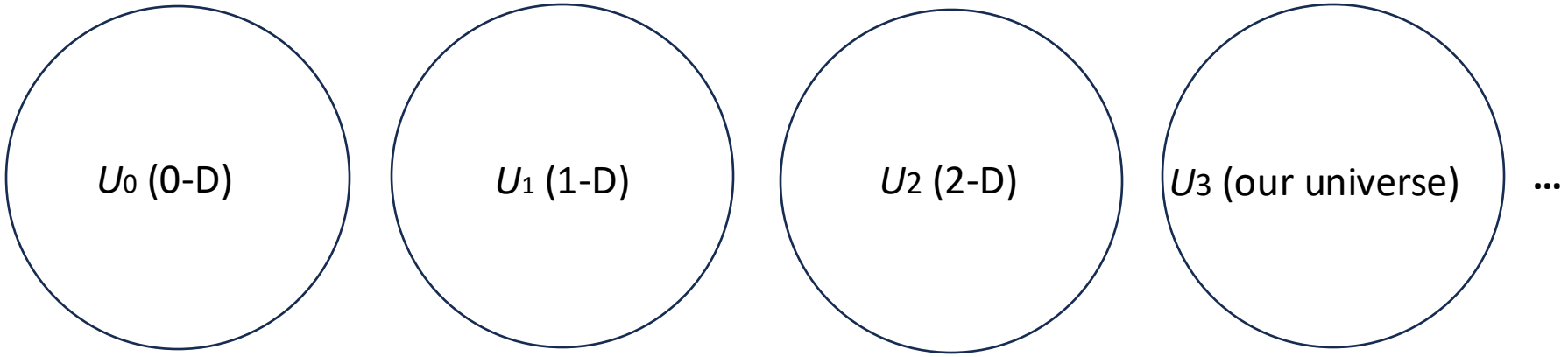
This model is far simpler and more reasonable than M-theory. However, we have never detected infinite dimensional space.



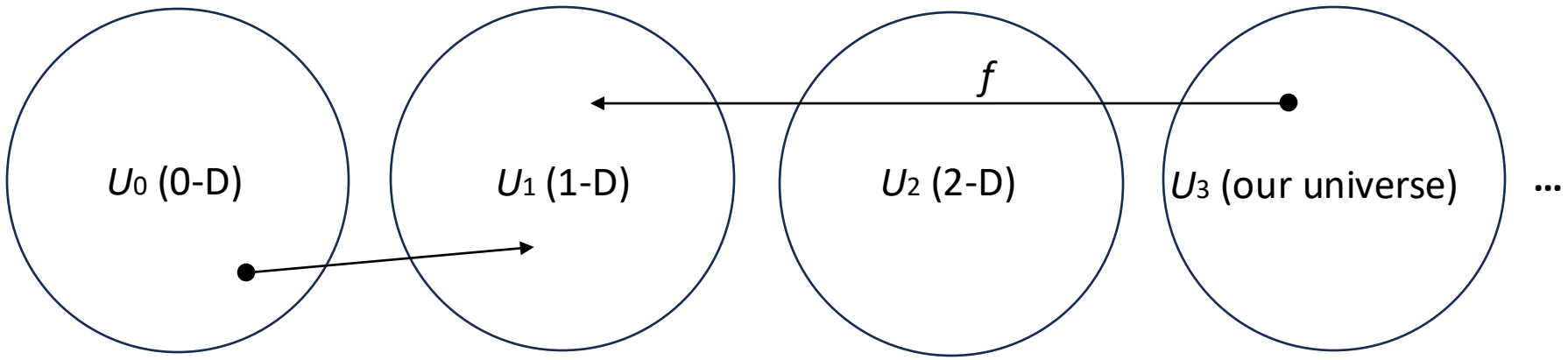
Non-empirical Thought of Dimensionality

We should not empirically think of dimensionality. We should not introduce U (universe in set theory) including U_n. Each dimensional space U_n is independent from others. There is neither interaction nor union. They are disjoint to each other.

$$\therefore U_0 \cap U_1 \cap U_3 \cap \cdots \cap U_n = (U_0 \cup U_1 \cup U_3 \cup \cdots \cup U_n)^c = \emptyset$$



In our theory of dimensionality, we think of a point mapped into another universe, U_n. However, for example, any point in U_3 cannot be a point in U_1 even after mapping f ($f : S \rightarrow S'$) because the point after the mapping should keep the character in U_3. Any point in U_n keeps its original n-degrees of freedom before and after mapping. Therefore, the changelessness of the degree of freedom is both invariant and symmetry as the degree of freedom is changeless after a series of operations.

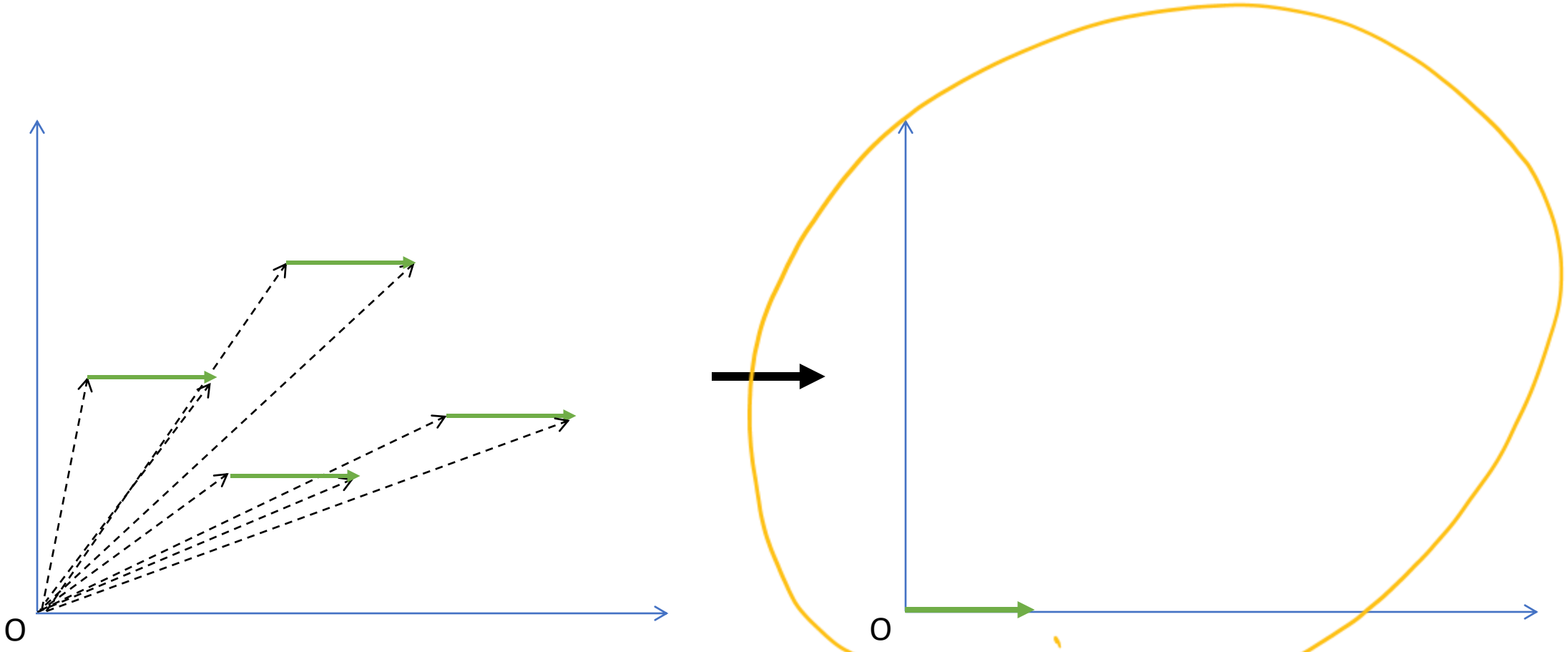


Furthermore,

Georg Cantor pointed out that since 1-dimensional set and 2-dimensional set have the same cardinality, dimensionality is nonsense. Giuseppe Peano was inspired by that idea and showed it graphically. It is known as 'Peano curve'. However, those ideas are wrong because they overlook the ‘freedom of degree’ that any point has in our discussion of dimensionality.

You may think that since vector has magnitude and direction, it clarifies dimensionality for the problem that Peano curve has. However, that is also wrong.

As shown in the graphs in the next slide, those vectors on the left-hand are identified as a vector on the right-hand. Those vectors are obviously in 2-dimensional space. Nevertheless, they are identifiable as a vector in the 1-dimensional graph.



The space by vector in the graph on the right-hand is known as 'quotient vector space'.

Conclusions

We are going to reconstruct geometry itself through our discussion of dimensionality, it is inevitable. It is therefore important also for mathematics, not only for physics.

We mainly reconstruct three classical geometries.

1. Euclidean geometry
2. Co-ordinate geometry by René Descartes and Pierre de Fermat
3. Erlangen Programme by Felix Klein

- For 1, any point could be a line and a plane in the orthogonal co-ordinates; as well as a circle and a sphere in the polar co-ordinates. There is also mapping from lower dimensional space into higher dimensional space. It contradicts the first to third postulate of ‘Elements’.

- For 2, any point keeps its original degree of freedom mapping into any other dimensional space. Descartes’s point is nothing more than a label in the co-ordinates. He got the idea of co-ordinate geometry when he saw a fly on a military tent. However, he overlooked the degree of freedom that the fly had.

- For 3, all geometries by mapping between different dimensional spaces are operated by a groupoid. Erlangen Programme is only about mapping in the same dimensional space. We could call it ‘mapping to itself’ because $E_{mm} = 1 = I$.

It is important not only for higher dimensional but also lower dimensional physics.

Furthermore,

- Difference between classical dimensionality and our discussion of dimensionality is similar to between geocentric theory and heliocentric theory. The Kaluza-Klein theory is based on a 3-dimensional centric idea, so that the extra dimensional space should be compactified.

- We could also say the difference is like between absolute space and relative space. Brane or braneworld belongs to ‘absolute’ higher or infinite dimensional space. In other words, any dimensional space belongs to higher dimensional space. However, our dimensionality is not like that. Any of different dimensional spaces are NOT a subspace of others. They are independent from each other.

Open Problems

Viewing topology from our theoretical viewpoint of dimensionality, we will need to redefine various concepts in topology.

For example, in topology, S^1 is equivalent to $\mathbf{R}\times\{\infty\}$. However, in the context of our theory of dimensionality, S^1 has two degrees of freedom in 2-dimensional space. Mapping S^1 to 1-dimensional space, it is $\mathbf{R}\times\{\infty\}$ having 'two degrees of freedom'. Therefore, 'degree of freedom' is invariant or symmetry from our viewpoint.

If mapping S^1 to R^3, S^1 becomes a cylinder. If S^1 to S^3, S^1 does T^2.

T^2 in S^3

Pullback should be redefined. For example, a point $(x, y) = (a, b)$ is mapped into 1-dimensional space then pull it back, it loses the information of y-coordinate, so it is $x = a$, a line in 2-dimensional space. However, in the context of our theory of dimensionality, it never loses the information because the two degrees of freedom is conserved because of its 'invariant' or 'symmetry'. Therefore, by our notation, $E_{21}E_{12} = E_{22} = I$. On the other hand, $x = a$ is a point in 1-dimensional space and line in 2-dimensional space because of its 'invariant'. Therefore,

$E_{12}E_{21} = E_{11} = I$. Though it is similar to pullback but it is based on quite a different concept of us.

Since merely introducing degrees of freedom as invariant in topology, Such the redefinition will be not a complicated issue.

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